Experiences of Teachers Using Project-Based Learning with Students with Emotional and Behavioral Disorders

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Dedication

This dissertation is dedicated to my grandfather, Albert Smeraldo, and my daughter, Lucy Sciacca.

Grandpa, thank you for instilling in me a love of learning, showing me how to approach life with curiosity and joy, and teaching me what it means to love unconditionally. You inspired my journey into education and reminded me that becoming a better person is a lifelong pursuit. Our family misses you every day.

Lucy, thank you for bringing that same spirit of discovery into my life in new ways. You remind me daily to stay curious, to have fun, and to see learning as an adventure. You help me remember why I chose this path and how learning allows us to grow together.

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Finally, thank you to my husband and daughter for giving me space and time to step away from my role as a wife and parent to do this incredibly rewarding work. I could not have done it without you.

Abstract

Students with emotional and behavioral disorders (EBD) are at risk for many adverse outcomes due to their complex needs (National Center for Education Statistics, 2024; Office of Special Education Programs, 2020; Smith et al., 2018). The difficulty teachers face in meeting those needs adds an additional challenge (Smith et al., 2018; Sutherland et al., 2019). Project-based learning is an instructional method that shares similarities with high-leverage practices (HLP) (Wertz & Mulcahy, 2024) and has been successful with students without disabilities (Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Gomez-Pablos et al., 2017; Revelle et al., 2020; Samsudin et al., 2020; Sormunen et al., 2019); however, there is little research on its use with students with EBD. This study aimed to understand the experiences and perceptions of teachers using PjBL with students with EBD. A collective case study was conducted using interviews and artifact data from three teachers using PjBL with students with EBD. Data from three participant interviews and artifacts were analyzed using HLP and the foundations of culturally inclusive pedagogies and practices as a theoretical framework. Findings highlighted the importance of building relationships with students to adapt to their needs, positive impacts across academic and non-academic domains, challenges related to planning and flexible grouping, and an overall positive effect on teachers and students. These insights offer valuable guidance for future research and practice focused on understanding and expanding the use of PjBL to better serve students with EBD.

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Chapter I. Statement of the Problem

Students with emotional and behavioral disabilities (EBD) face challenges in traditional educational environments. They are challenged socially, emotionally, and behaviorally and struggle academically (Chen et al., 2021; Smith et al., 2018). These challenges lead to adverse outcomes, including negative impacts on academic achievement and behaviors (Chen et al., 2021; Smith et al., 2018), increased rates of high school dropouts (National Center for Education Statistics, 2024), and isolation from their typical, same-aged peers (Office of Special Education Programs, 2020). Implementing evidence-based practices with these students can be difficult because of their complex needs (Farley et al., 2023; Smith et al., 2018; Sutherland et al., 2019). One evidence-based intervention might work for some students with EBD, but the multiple variables that influence a student's behaviors and achievement require individualized adaptive approaches (Farley et al., 2023).

It is important to identify instructional approaches that teachers successfully implement with a positive impact on their students. Project-based learning (PjBL) is an approach that has had a positive impact on the academic and social-emotional needs of non-disabled students (Culclasure et al., 2019; Febriana, 2017; Lee et al., 2015; Samsudin et al., 2020). While this approach has limited research on students with EBD (Taylor et al., 2024), the positive impact it has on students with learning disabilities (Berkeley et al., 2019; Lu et al., 2022; Sormunen et al., 2019) and visually impaired, deaf, or hard of hearing (Eldiva & Azizah, 2019) may be replicated across populations. This case study explored the experiences and perceptions of teachers who have used PjBL with students with EBD, focusing on their views regarding the impact on students and the implementation process.

Problem Statement

The problem is that students with EBD have complex needs. The multifaceted challenges students with EBD face have adverse outcomes related to academics and behavior (Chen et al., 2021; Smith et al., 2018) and cause disengagement (Didion et al., 2018). While the national average dropout rate for all students with disabilities in the 2021-2022 school year was 15%, 39% of students with emotional disturbance dropped out of high school (National Center for Education Statistics, 2024), an increase from 32% in the 2017-2018 school year (Office of Special Education Programs, 2020). High school dropouts are more likely to be unemployed or receive low wages (U.S. Bureau of Labor Statistics, 2024). Individuals without a high school diploma have a 5.6% unemployment rate, compared with 3.9% for individuals with a high school diploma. Dropouts earn a median weekly income of $708, compared with $899 for high school graduates. (U.S. Bureau of Labor Statistics, 2024). In 2023, the average monthly expenses in the United States totaled $6,440 (U.S. Bureau of Labor Statistics, 2023), nearly double the income of a high school dropout. Earning an average annual salary of $36,816 (U.S. Bureau of Labor Statistics, 2024) places them far below the average cost of living in Pennsylvania, which was $87,500 per year for a family of four (Adams, 2024).

Students with disabilities represent 17% of the K-12 population but account for 24% of in-school suspensions, 29% of out-of-school suspensions, 21% of expulsions, and 28% of school-related arrests (U.S. Department of Education Office of Civil Rights, 2023). Exclusionary discipline practices lead to lower academic achievement (Wang et al., 2022), decreased student engagement (Cohen et al., 2023; Wang et al., 2022), increases in disruptive behavior, adverse impacts on social and emotional skills (Cohen et al., 2023), and an increase in arrests (Novak, 2022).

Students with disabilities also accounted for 81% of physical restraints, 32% of mechanical restraints, and 75% of seclusionary practices (U.S. Department of Education Office of Civil Rights, 2023). Seclusionary practices encompass any "involuntary confinement of a student alone in any room or area. It includes the use of any room or area in which the student is alone and not free to leave (or believes they are not free to leave)" (U.S. Department of Justice, 2024, para. 1). Students with EBD had more than twice the number of disciplinary removals as compared to other students with disabilities (Office of Special Education Programs, 2020). Students with EBD are more likely to attend school in a separate setting, with 12% attending a separate school as compared to 3% of all students with disabilities (Office of Special Education Programs, 2020). Students with EBD are overrepresented in the juvenile justice system and more likely to recidivate (Kim et al., 2022). Their academic, social, emotional, and behavioral challenges can cause decreased student engagement and poor social skills (Smith et al., 2018) that contribute to these adverse outcomes (National Center for Education Statistics, 2024; Office of Special Education Programs, 2020; Smith et al., 2018).

Though research-based practices are available, it has been difficult to implement those practices in schools due to the complex needs of students with EBD (Farley et al., 2023; Smith et al., 2018; Sutherland et al., 2019). Teachers often feel unprepared to meet the students' needs and struggle to implement "effective instructional and classroom management strategies" (Gilmour et al., 2022, p. 16). Lack of knowledge and negative attitudes toward students with EBD present an additional challenge (Alkahtani, 2022; Gilmour et al., 2022) that reduces teachers' ability to implement academic and non-academic strategies with fidelity (Gilmour et al., 2022). The stresses of teaching students with EBD can lead to educator burnout (Gilmour et al., 2022; Hirsch et al., 2023) and an increased risk of turnover for both special educators and non-special

educators (Gilmour & Wehby, 2020). Understanding teachers' experiences who use PjBL with students with EBD can help inform practices relative to these students' complex academic, behavioral, social, and emotional needs.

Purpose of the Study

Identifying and implementing the most effective strategies to use when teaching students with EBD remains a challenge based on the individual needs of each student (Criss et al., 2024; Gilmour et al., 2022; Scheibel, 2023; Sutherland et al., 2019). Project-based learning (PjBL) offers a way to engage students with EBD academically while addressing social-emotional learning academically (Culclasure et al., 2019; Febriana, 2017; Lee et al., 2015; Samsudin et al., 2020). PjBL is a constructivist approach (Jumaat et al., 2017) that promotes active learning (Gomez-Pablos et al., 2017; Sormunen et al., 2019) with positive effects on social and emotional learning (Culclasure et al., 2019; Febriana, 2017; Lee et al., 2015; Samsudin et al., 2020). Studies have shown positive effects on students in areas of participation (Gomez-Pablos et al., 2017; Sormunen et al., 2019), critical thinking, and other academic skills (Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Revelle et al., 2020). PjBL is linked to positive outcomes with cooperation (Shin, 2018; Sormunen et al., 2019), motivation (Dole et al., 2017; Duke et al., 2021; Lee et al., 2015; Sari, 2018; Shin, 2018), and engagement (Dole et al., 2017; Gomez-Pablos et al., 2017). Most research involves PjBL with non-disabled students (Almulla, 2020; Dole et al., 2017; Gomez-Pablos et al., 2017; Sormunen et al., 2019), and studies that did include students with disabilities focused on those with speech, reading, and writing disabilities (Berkeley et al., 2019), learning disabilities (Lu et al., 2022; Sormunen et al., 2019), and students who were blind, deaf, or intellectually disabled (Eldiva & Azizah, 2019).

PjBL could be an educational practice teachers can implement that may have positive effects academically, behaviorally, socially, and emotionally for students with EBD, based on the research with students who were non-disabled (Almulla, 2020; Culclasure et al., 2019; Dole et al., 2017; Duke et al., 2021; Febriana, 2017; Gomez-Pablos et al., 2017; Lee et al., 2015; Revelle et al., 2020; Samsudin et al., 2020; Sari, 2018; Shin, 2018) in inclusive settings (Sormunen et al., 2019), and self-contained classrooms (Taylor et al., 2024).

The purpose of this study was to understand the experiences and perceptions of teachers using PjBL with students with EBD using a collective case study design. The benefits of using this approach were that it allowed me to understand and report the experiences of individual teachers implementing PjBL with students with EBD. A collective case study design allowed me to gain multiple perspectives (Creswell & Poth, 2018) around the implementation of PjBL with this population. The population was selected through criterion sampling. Participants were educators who taught using PjBL in classes with students with an IEP disability category of emotional disturbance. This included a special educator in a self-contained setting, a special educator who taught in both a self-contained setting and supported general education teachers, and a general educator with inclusive classrooms. The grade levels targeted included students in grades 6-12.

This approach allowed me to gather relevant and timely data (Creswell & Poth, 2018) on educators currently using PjBL. I collected multiple forms of data to understand their perceptions through interviews and their implementation process through artifacts, and teacher-created documents (Creswell & Poth, 2018). It was important to understand the specific characteristics of the experiences and how they were implemented to analyze teachers' perceptions of how PjBL impacted their students and their views on the effectiveness of employing it as an

instructional methodology. A collective case study focusing on the implementation of PjBL with students with EBD offered the opportunity to understand teachers' insights regarding the perceived effectiveness and feasibility of PjBL with students with EBD.

Research Question

Considering the challenge teachers face in meeting the complex needs of students with EBD, the research question was: What are the experiences and perceptions of teachers who use project-based learning with students with emotional and behavioral disorders?

Theoretical Framework

Teachers of students with disabilities have access to different research-based instructional practices; however, it can be challenging to find the time to identify, study, and practice the different research-based strategies (Brownell et al., 2021). In response to this challenge, the Council for Exceptional Children (CEC) and the Collaboration for Effective Educator Development, Accountability, and Reform (CEEDAR) Center synthesized 22 high-leverage practices (HLPs) called the High Leverage Practices for Students with Disabilities (Aceves & Kennedy, 2024). These practices can and should be used by any educator who works with students with disabilities, including those teaching in general education classrooms (Aceves & Kennedy, 2024).

The HLPs are rooted in culturally inclusive pedagogies and practices (CIPP) (Taylor & Monaco, 2024). Educators can more effectively address all students' academic, behavioral, social, and emotional needs by being more culturally aware and inclusive. CIPP guides educators to see students as "complex individuals" (Taylor & Monaco, 2024, p. 19) with "multilayered identities, experiences, and cultures" (Taylor & Monaco, 2024, p. 19). These guiding ideas help educators to understand and address the complex, individualized needs of students with EBD

(Smith et al., 2018; Sutherland et al., 2019) by understanding that the needs of these students are also intertwined with their identities, cultures, and environments (Taylor & Monaco, 2024). HLP and its theoretical underpinnings guided data collection and analysis by focusing on the teacher's experiences, perceptions, and implementation of PjBL with students with EBD.

Significance of the Study

This study sought to address the gap in research by investigating the use of PjBL with students with EBD. There is minimal research on the use of PjBL and students with EBD. One paper suggested using the arts with PjBL to support students with challenging behaviors, arguing that art-based PjBL allows for more opportunities for student expression outside typical academic contexts (Taylor et al., 2018). Other research found that PjBL positively affected behaviors and teacher burnout, which is where the current research ends (Taylor et al., 2024).

Researchers state the importance of investigating ways to engage students with EBD (Didion et al., 2018; Taylor et al., 2024) and the effects of PjBL on students with disabilities (Dole et al., 2017; Taylor et al., 2018; Taylor et al., 2024). It is important to understand whether these positive effects carry over to students with EBD to address the challenge of effectively educating them and reducing the likelihood of adverse outcomes (Bettini et al., 2016; Smith et al., 2018; Sutherland et al., 2019). Students with EBD could benefit from this research as it may lead to understanding how PjBL impacts their learning outcomes.

Finding solutions to engagement and achievement for students with EBD is important for both the students and the field of education (Bettini et al., 2016). The findings of this study could influence general and special educators working with students with EBD. Identifying instructional methods that can be used with high-leverage practices for the highest impact on

students with EBD would benefit teachers and the students in inclusive and alternative education settings.

Other stakeholders that may benefit from this research include educational leaders, such as curriculum directors, special education directors, and building-level principals. If these individuals are made aware of the possible benefits of PjBL for students with EBD and how teachers implement PjBL, they can use this knowledge to ensure their teachers have the appropriate support and knowledge to use it in their settings.

Definitions of Terms

The following terms are defined to help understand the content of this study.

Academic Needs—Areas in which students with EBD require support or intervention to attain grade-level academic skills.

Asset-based approach—An approach to teaching that focuses on the strengths and talents a student brings and how those strengths and talents can be utilized to support the learning of each student (NYU Steinhardt, 2018).

Behavioral Needs—Areas in which students with EBD require support or intervention to attain developmentally appropriate behaviors.

Culturally Inclusive Pedagogies and Practices (CIPP)—Teaching approaches that guide educators to be aware of cultural differences, inclusive of the different backgrounds of their students, and supporting these differences in their instructional practices and relationships with students (Taylor & Monaco, 2024).

Culturally Responsive Pedagogy— A teaching approach that embeds the awareness of students' social and cultural backgrounds and how these factors impact their success (Ladson-Billings, 1995).

Culturally Relevant Pedagogy—A teaching approach that guides teachers to build awareness and be supportive of the cultural identities of their students (Brown-Jeffy & Cooper, 2011).

Culturally Sustaining Pedagogy—A teaching approach that guides teachers to support students' cultural and linguistic differences (Paris, 2012).

Emotional and Behavioral Disabilities (EBD)—"A variety of conditions and symptoms that impair learning, social functioning, safety, and behavioral health of children or adolescents" (Vanderbilt Kennedy Center, 2014, p. 1).

Emotional and Behavioral Disorders—"A variety of conditions and symptoms that impair the learning, social functioning, safety, and behavioral health of children or adolescents" (Vanderbilt Kennedy Center, 2014, p. 1).

Emotional Disturbance—

> (i) Emotional disturbance means a condition exhibiting one or more of the following characteristics over a long period of time and to a marked degree that adversely affects a child's educational performance:
>
> > (A) An inability to learn that cannot be explained by intellectual, sensory, or health factors.
> >
> > (B) An inability to build or maintain satisfactory interpersonal relationships with peers and teachers.
> >
> > (C) Inappropriate types of behavior or feelings under normal circumstances.
> >
> > (D) A general pervasive mood of unhappiness or depression.
> >
> > (E) A tendency to develop physical symptoms or fears associated with personal or school problems.

> (ii) Emotional disturbance includes schizophrenia. The term does not apply to children who are socially maladjusted unless it is determined that they have an emotional disturbance. (Individuals with Disabilities Act, 2004, § 300.8)

Inclusive Classrooms—Classrooms in which students with disabilities are educated alongside their non-disabled peers (Reading Rockets, 2024).

Individualized Education Program (IEP)—"A written plan for the provision of services for the education of students who are disabled or gifted" (Commonwealth of Pennsylvania, 2024).

Intersectionality—A framework that focuses on the interconnectedness of multi-level identities and how these identities shape individuals' experiences, including their privileges and experiences with oppression (Garcia, 2024).

High Leverage Practices (HLP)—Practices identified as effective when used with students with disabilities to improve their instruction and behavior (Brownell et al., 2021).

High Quality Project Based Learning (HQPBL)—A framework outlining six criteria that identify what high quality project based learning experiences should include for students (The Buck Institute of Education, 2018).

Multi-tiered system of supports (MTSS)—"A proactive and preventative approach that integrates data and instruction to maximize student achievement and support students' social, emotional, and behavioral needs from a strengths-based perspective" (American Institutes for Research, 2024a, para. 1).

Project-Based Learning (PjBL)—" a teaching method in which students gain knowledge and skills by working for an extended period to investigate and respond to an authentic, engaging, and complex question, problem, or challenge" (PBLWorks, 2024d, para. 3).

Replacement Behaviors— “a positive alternative behavior that serves the same function” (Iris Center, 2025b, para. 1).

Seclusionary Practices—Any "involuntary confinement of a student alone in any room or area. It includes the use of any room or area in which the student is alone and not free to leave (or believes they are not free to leave) (U.S. Department of Justice, 2024, para. 1).

Self-contained classrooms—A special education placement where special education teachers instruct a class that includes only students with disabilities (Spencer, 2013).

Social Emotional and Learning (SEL)—

> The process through which all young people and adults acquire and apply the knowledge, skills, and attitudes to develop healthy identities, manage emotions and achieve personal and collective goals, feel and show empathy for others, establish and maintain supportive relationships, and make responsible and caring decisions. (CASEL, 2024, para. 1)

Social and Emotional Needs—Areas in which students with EBD require support or intervention to attain developmentally appropriate social and emotional skills.

Summary

Students with EBD are at risk for several adverse outcomes due to their complex needs (Chen et al., 2021; National Center for Education Statistics, 2024; Office of Special Education Programs, 2020; Smith et al., 2018), and the difficulty teachers face in meeting those needs adds an additional challenge (Farley et al., 2023; Smith et al., 2018; Sutherland et al., 2019). PjBL is an instructional approach that has shown positive outcomes with non-disabled students (Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Gomez-Pablos et al., 2017; Lee et al., 2015; Revelle et al., 2020; Samsudin et al., 2020; Shin, 2018; Sormunen et al., 2019). There is little research on its use with students with EBD. This collective

case study sought to understand the experiences and perspectives of teachers using PjBL with students with EBD.

Chapter II. Review of the Literature

This collective case study examined the experiences and perspectives of teachers using project-based learning (PjBL) with students with emotional and behavioral disorders (EBD). The literature review begins by focusing on the background of students with EBD and current strategies utilized to address their multifaceted needs. High-leverage practices were applied as a framework as the review synthesizes the research on PjBL and its impact on students with EBD and their teachers.

Students with EBD

Emotional and behavioral disorders are defined as "a variety of conditions and symptoms that impair the learning, social functioning, safety, and behavioral health of children or adolescents" (Vanderbilt Kennedy Center, 2014, p. 1). Characteristics of individuals with EBD can include hyperactivity, aggression, self-injurious behavior, withdrawal, immaturity, and learning difficulties (Division for Emotional and Behavioral Health, 2024; Kauffman & Landrum, 2018; Vanderbilt Kennedy Center, 2014). The most severe individuals "may exhibit distorted thinking, excessive anxiety, bizarre motor acts, and abnormal mood swings" (Division for Emotional and Behavioral Health, 2024, para. 8). For educational services, these students are often identified as having an emotional disturbance (ED) (Individuals with Disabilities Act, 2004).

History of Emotional Disturbance Definition

The term emotional disturbance was first formally used in the early 1900s for students who were "incorrigible and truants" (Wills-Jackson, 2019, p. 82). The term seriously emotionally disturbed was added in 1963 with the passing of Public Law 88.164 (Mental Retardation Facilities and Community Health Centers Act, 1963) and again used in 1975 with the passage of

The Education for All Handicapped Children Act of 1975. This definition did not encompass the socially maladjusted (Wills-Jackson, 2019). When this act was reauthorized in 1997 as IDEA, the term was officially changed to emotional disturbance (ED) (Individuals with Disabilities Education Act, 2004).

In education today, students are diagnosed with emotional disturbance under the Individuals with Disabilities Education Act (Individuals with Disabilities Education Act, 2004). Qualification for an educational diagnosis of emotional disturbance is determined by meeting one or more criteria:

(A) An inability to learn that cannot be explained by intellectual, sensory, or health factors.

(B) An inability to build or maintain satisfactory interpersonal relationships with peers and teachers.

(C) Inappropriate types of behaviors or feelings under normal circumstances.

(D) A general pervasive mood of unhappiness or depression

(E) A tendency to develop physical symptoms or fears associated with personal or school problems. (Individuals with Disabilities Education Act, 2004)

Challenges with the Definition

The federal definition of ED continues to create challenges when considering how to best serve this population of students (Freeman et al., 2019). It excludes students with social maladjustments and is not objective in identifying the amount of time a student must demonstrate the criteria to be identified and what constitutes inappropriate behavior (Freeman et al., 2019; Kavale et al., 2004). There are similar challenges for the definition of learning disability, which also focuses on an inability to learn that "cannot be explained by intellectual, sensory, or health

factors" (Individuals with Disabilities Education Act, 2004). Often, students with learning disabilities also exhibit behavioral challenges (Kavale et al., 2004). As students with EBD often are academically below grade level, the question becomes whether these deficits are related to a learning disability or if their behaviors prevent them from learning at a typical pace. The complex needs of students with EBD also present a challenge in diagnosis and treatment, as comorbidities are often not recognized (Kavale et al., 2004).

The criteria for identifying a student with ED have remained unchanged since it was codified into law, unlike learning disabilities, which were updated to reflect current practice and research (Hanchon & Allen, 2017). The IDEA definition outlines that all the criteria must be met, but fails to outline the measurement of these criteria (Mitchell et al., 2019). This definition has posed a problem for school psychologists attempting to identify students appropriately when evaluating them for special education services, relying on the subjectivity of the evaluator. The subjective nature of evaluation contributes disproportionality in identifying students with externalized behaviors and gender and race inequities (Hanchon & Allen, 2017).

The Council for Children with Behavioral Disorders advocated changing the label of serious emotional disturbance, as it attached a stigma to individuals with these disabilities and adopted "the terminology Emotional and Behavior Disorder" (Wills-Jackson, 2019, p. 83). Fourteen states have or will officially change from emotional disturbance to emotional disability to help mitigate this stigma (Arundel, 2024; Monserrate, 2024; Salhotra, 2021). Following their work, the National Mental Health and Special Education Coalition further outlined a more precise definition for EBD (Mitchell et al., 2019). They defined students with EBD as:

> (i) The term emotional or behavioral disorder means a disability characterized by behavioral or emotional responses in school programs so different from appropriate age,

> cultural, or ethnic norm that the responses adversely affect educational performance, including academic, social, vocational, and personal skills. Such a disability
>
> > (A) is more than a temporary, expected response to stressful events in the environment.
> >
> > (B) is consistently exhibited in two different settings, at least one of which is school-related; and
> >
> > (C) is unresponsive to direct intervention in general education, or the child's condition is such that general education interventions would be insufficient.
> >
> > > (ii) Emotional and behavioral disorders can co-exist with other disabilities.
> > >
> > > (iii) This category may include children or youth with schizophrenic disorders, affective disorder, anxiety disorder, or other sustained disorders of conduct or adjustment where they adversely affect educational performance in accordance with section (i) (Forness & Knitzer, 1992, p. 13)

This definition addressed some issues around concurrent disabilities and differences in culture and ethnicities (Kavale et al., 2004). It clarified who would be eligible, provided a multi-step evaluation process, and used multiple forms of data across settings (Mitchell et al., 2019). The definition also expands upon the educational effects "to include academic, social, vocational, and personal skills" (Mitchell et al., 2019, p. 79). It has yet to be included in IDEA.

In Pennsylvania, the disability category does not automatically determine the types of supports they can receive (22 PA Code Chapter 14 § 14.131). "Students may receive more than one type of support as appropriate" (22 PA Code Chapter 14 § 14.131, para 1). Students with EBD may be identified under different primary disability categories in their IEP, such as ED,

other health impairment (OHI), intellectual disabilities (ID), or autism. For this study, I focused specifically on teachers with students whose primary disability category was ED or a combination of ED and OHI to minimize potential data inconsistencies and maintain a clear focus on a defined group. The term EBD is used throughout this report to refer to students who were identified with ED or ED and OHI in school settings, received emotional support services, had a positive behavior support plan included in their IEP, or whose IEP includes services and supports related to social or emotional skill development or functional behavior (22 PA Code Chapter 14 § 14.131, para. iv). EBD is also a more inclusive and less stigmatizing way to describe these students and is used by other researchers in the field (Culclasure et al., 2019; Kumm et al., 2021).

Challenges and Adverse Outcomes

"Students with EBD display a wide range of needs, but common characteristics include poor social skills acquisition, disruptive behavior, and academic performance deficits" (Kumm et al., 2021, p. 97). Students display behaviors such as inattentiveness, disruptive behaviors, and rule-breaking. These behaviors can cause students with EBD to enter a failure cycle (Sanders et al., 2018). This cycle begins with students displaying inappropriate behaviors, which prompts them to be removed from the classroom. Their removal leads to lost instructional time, negatively impacting their academic achievement. Lost instructional time and decreased academic skills continue to impact their behavior negatively, and the cycle continues. Learning loss negatively impacts present and future achievement (Frantz, 2023). These complex challenges negatively impact their academic and behavioral outcomes (Ennis et al., 2018).

Students with EBD often have deficits in all academic areas (Kumm et al., 2021; Ennis et al., 2018) and perform one or two grade levels below their peers (Freeman et al., 2019). They

struggle with reading (Mathews et al., 2021) and written expression (Evmenova & Regan, 2019). Students demonstrate decreased achievement in literacy and numeracy and are at risk of academic failure (Vanderbilt Kennedy Center, 2014). They have poor social skills and engage in disruptive behaviors that prompt them to be removed from the classroom environment (Kumm et al., 2021; Hutchins et al., 2020). These challenges lead to a higher dropout risk (Freeman et al., 2019; Vanderbilt Kennedy Center, 2014).

Students with EBD display social skills deficits in relationships, self-management, academic skills, compliance, and assertion (Hutchins et al., 2020). Social skill deficits lead to adverse outcomes in relationships. Students with EBD report higher rates of bullying and are less happy at school compared with other students with different disabilities (Lipscomb et al., 2017).

Students with EBD are one of "the most socioeconomically disadvantaged groups and the most likely to attend lower-performing schools" (Lipscomb et al., 2017, p. i). A longitudinal study found that 62% of students with EBD lived in low-income households, and 33% attended a low-performing school (Lipscomb et al., 2017). Students with EBD are less likely to have parental assistance with homework (54%) compared to the overall population of students with IEPs (72%). Additionally, they show lower participation in school events, with only 46% engaged versus 58% of all students with IEPs.

These challenges can lead to negative consequences beyond the educational system (Chen et al., 2021; National Center for Education Statistics, 2024; Offerman et al., 2022; Office of Special Education Programs, 2020; U.S. Department of Education, 2023; Vanderbilt Kennedy Center, 2014). Students with EBD face higher rates of adverse outcomes (Chen et al., 2021) and begin their education with traumatic experiences that negatively impact their future (Offerman et al., 2022). Adverse childhood experiences (ACEs) are traumatic events that a child experiences

between the ages of 0-17 (CDC, 2024). While approximately 64% of adults report having experienced at least one ACE as a child, a study of students with EBD in a special education school found that 96.4% had experienced at least one adverse childhood experience (ACE), and more than half experienced four or more (Offerman et al., 2022). Children who experience ACEs are more likely to develop emotional and behavioral problems, along with other negative outcomes. These include lower academic achievement, higher rates of dropout and arrest, and negative physical effects. Students who experience ACEs are also at higher risk of heart disease, depression, and suicide (CDC, 2024).

Among all students with disabilities, those identified with EBD had the highest dropout rate in 2021, with 28% leaving school before graduation (National Center for Education Statistics, 2024; Office of Special Education Programs, 2020; U.S. Department of Education, 2023). Students with EBD are likely to be served outside the regular education classroom (Freeman et al., 2019). The U.S. Department of Education (2023) reported the following regarding the educational environments:

Table 1

Percentage of Time in Regular Education Classrooms for Students with Disabilities in 2021

Population	80% or more	40% to 79%	Less than 40%	Other settings
Students with EBD	54.7%	17%	14.6%	13.7%
All students with disabilities	66.7%	16%	12.5%	2.4%

Students with EBD face significantly higher rates of disciplinary actions compared to other students with disabilities (Lipscomb et al., 2017). Specifically, 65% of students with EBD experience suspension, more than double the rate of 29% for all students with IEPs. Nineteen percent of students with EBD face expulsion, which is over twice the rate of 8% observed in the

overall population of students with IEPs (Lipscomb et al., 2017). Students with EBD in self-contained settings experience the highest rates of out-of-school suspension, expulsion, and arrests among all disability categories (Gersib & Mason, 2023). In the 2020-2021 school year, 55 out of every 10,000 students with EBD received OSS or expulsions for more than 10 days, and 29 out of every 10,000 received in-school suspensions (U.S. Department of Education, 2023). Students identified with other disabilities have a rate of 15 or less per 10,000. Higher rates of exclusionary practices can lead to "more concentration problems, disruptive behavior, and emotion dysregulation, and less prosocial behavior" (Cohen et al., 2023, p. 9), decreased engagement and academic achievement (Wang et al., 2022), and an increased risk of dropout or arrest (Cohen et al., 2023). Students with EBD are more frequently placed in alternative education settings than those with any other disability (Freeman et al., 2019). The placement rate for students with EBD is 15 in every 10,000, which is significantly higher than the rate of six out of 10,000 for students with other disabilities (U.S. Department of Education, 2023).

Students with EBD experience a significantly higher risk of arrest and incarceration (Freeman et al., 2019; Gersib & Mason, 2023). Research shows that 17% of students with EBD experience arrest, nearly triple the rate of six percent of all youth with disabilities (Lipscomb et al., 2017). Given the magnitude of these issues, it is important to examine how these difficulties affect not only the students but also the impact on educators working with this population.

Impact on Educators

Special education teachers (SETs) who work with students with EBD are at a high risk of burnout (Brunsting et al., 2022; Brunsting et al., 2023). These teachers report higher levels of emotional exhaustion than other teachers (Brunsting et al., 2022). Their risk for burnout and exhaustion only amplifies the risk of lower use of evidence-based practices with fidelity.

Students of teachers experiencing burnout experience negative impacts on their motivation (Shen et al., 2015), ability to meet goals (Wong et al., 2017), and decreased academic achievement (Madigan & Curran, 2021). Student motivation is influenced by the level of burnout teachers experience in general education settings (Shen et al., 2015). There are statistically significant relationships between levels of teacher stress and teaching quality ($p < .001$) and student engagement ($p = .006$) (Wong et al., 2017). Teachers' sense of accomplishment significantly predicted IEP outcomes ($p = .005$) (Wong et al., 2017). Teachers with less support (including lack of a partner teacher, lack of paraprofessionals, and inadequate time) provide weaker instruction to their students (Mathews et al., 2021). Teacher burnout negatively impacts student exam scores, grades, and GPA (Madigan & Curran, 2021).

Despite these negative impacts, SETs of students with EBD report higher levels of personal achievement and lower levels of depersonalization than other primary or secondary teachers (Brunsting et al., 2022). Special education teachers of students with EBD are more likely to have reduced burnout if they feel supported by their administrators, access to curricular resources, and perceive their workload to be manageable with enough planning time (Bettini et al., 2020; Brunsting et al., 2023; Cumming et al., 2021). In these conditions, teachers report higher use of evidence-based practices with their students (Cumming et al., 2021). It is important for SETs to have a manageable workload and support not only to prevent burnout and lower stress but also to improve their effectiveness in teaching students with EBD (Bettini et al., 2020; Brunsting et al., 2023; Cumming et al., 2021).

Approaches to Support Students with EBD

In 1991, the Peacock Hill Working Group (1991) published their findings on effective practices for teaching students with EBD. During that time, less than half of the students with

EBD were in inclusive settings, with a dropout rate of 40%. Teacher shortages for this population were prevalent, with high attrition rates. They recommended successful strategies and programs, including systematic data-based interventions, continuous assessment and monitoring of progress, provision for the practice of new skills, treatment matched to the problem, multi-component treatment, programming for transfer and maintenance, and commitment to sustained intervention. Programs they identified that were successful included preventative programs in preschool that included parent education and support, elementary programs that included aspects of peer modeling, explicit social skills curriculum, classroom contingency programs, and token economy systems. Secondary programs included token economy systems and social skills instruction. Following their research, they recommended preventative policies and practices and increasing funding for programs and policies focused on cross-agency collaboration. While these recommendations gained some traction, such as Part D of IDEA funding research for effective funding for programs (Freeman et al., 2019), the present risks for students with EBD remain high (Chen et al., 2021; National Center for Education Statistics, 2024; Office of Special Education Programs, 2020).

Effective strategies that have been empirically tested include:

> (a) teacher praise (reinforcement); (b) high rates of opportunities to respond; (c) precision requests and behavioral momentum; (d) clear instructional methods, including direct instruction, class-wide peer tutoring, and continuous monitoring of student performance (e.g., curriculum-based measurement); (e) positive behavior support, including functional assessment-based planning, self-management, and modifying antecedent and consequence events; and (f) group-oriented contingencies. (Mitchell et al., 2019, p. 78)

Tiered systems of support allow educators to implement evidence-based problem-solving and tools to positively impact all students (Mitchell et al., 2019). "A multi-tiered system of supports (MTSS) is a proactive and preventative approach that integrates data and instruction to maximize student achievement and support students' social, emotional, and behavioral needs from a strengths-based perspective" (American Institutes for Research, 2024a, para. 1). There are three tiers in MTSS (American Institutes for Research, 2024b). Tier 1 strategies are those that can be helpful to all students. Tier 2 strategies focus on small group interventions for students who are identified as being at-risk. Tier 3 strategies are for individuals who do not respond to Tier 1 or Tier 2 strategies or require more intensive, individualized intervention. Teachers focus on Tier 3 strategies for individuals with significant EBD.

A meta-analysis of behavioral interventions in 2022 outlined eight evidence-based practices for students with EBD: Check-in check-out (CICO), functional assessment-based interventions, group contingencies, peer-mediated interventions, self-management, self-regulated strategy development, token economies, and video modeling (Riden et al., 2022). In addition, students with EBD require social skills interventions (Kumm et al., 2021), which allow these students to develop positive relationships with others and learn more effectively. Other strategies showed promise, such as "praise, OTR [opportunities to respond], behavior contracting, cooperative learning, goal setting, good behavior game, high probability requests, instructional choice, self-determination, social skills, and time out" (Riden et al., 2022, p.262).

Inclusive Settings

The evidence of effective practice for students with EBD in inclusive settings remains limited (Soares et al., 2024). Current research lacks empirically supported recommendations for this population. A meta-analysis in 2024 found three strategies with the potential for positive

outcomes for students with EBD who spend their time in inclusive settings: a self-determined learning model of instruction (Kelly & Shogren, 2014), behavior-specific praise (BSP) (Allday et al., 2012), and the use of sound amplification (Maag & Anderson, 2006).

A self-determined model of instruction provides students with skills in making choices, self-advocating, self-determination, and self-regulation (Kelly & Shogren, 2014). This model allows the teacher to implement it in any curricular area to apply the skills to a student-set goal. When implemented, this intervention improved and maintained on-task behaviors for high school students with EBD.

Behavior-specific praise (BSP) directs teachers to write or verbally give students specific positive feedback, and the impact of the behavior of students on controllable behaviors, such as "Good job studying for this science test, your effort paid off" (Royer et al., 2019, p. 113). Students with EBD in inclusive settings have experienced positive outcomes when exposed to BSP (Allday et al., 2012), consistent with findings in other settings (et al., 2019).

Sound amplification addressed attention problems with students with EBD (Maag & Anderson, 2006). The amplified sound increased the rate at which students responded to low-interest directions. This intervention did not impact the rate at which students responded to high-interest directions.

Self-Contained Settings

A self-contained classroom is a special education placement where special education teachers instruct a class that includes only students with disabilities (Spencer, 2013). A meta-analysis in 2023 found six themes consistent with research practices for students with EBD in elementary self-contained settings (Gersib & Mason, 2023). Themes included "(1) a highly structured classroom environment, (2) use of positive strategies, (3) explicit social-emotional

(SEL) instruction, (4) applied behavior strategies, (5) classroom engagement, and (6) maintaining a relational focus" (Gersib & Mason, 2023, p. 273). A highly structured environment with clear rules and expectations reduced negative behaviors. Positive strategies rather than punitive strategies had a greater impact on student behavior. Social and emotional skills must be explicitly taught and reinforced. Skills include, but are not limited to, problem-solving, emotional regulation, social skills, behavior rehearsal, and self-management. Applied behavior strategies such as modeling, instructive feedback, and error correction improve social outcomes. Classroom engagement strategies included opportunities to respond, active supervision, academic engagement, instructional feedback, and scaffolding. The relational component showed that the stronger the teacher-student relationship, the lower the number of aggressive behaviors. Of these six themes, classroom structure and positive strategies had a positive, though nonsignificant impact, and "SEL, applied behavior techniques, and engagement resulted in negative estimates" (Gersib & Mason, 2023, p. 278), with recommendations for further research in self-contained settings.

Other Strategies

Schools must ensure their programs are aligned with academic standards with appropriate "modifications, adaptations, and accommodations" (PaTTAN, 2023, p.1) and provide social and emotional support and behavior management based on student needs. Due to the complex needs of students with EBD, researchers recommend that teachers use a toolbox method, utilizing common practice elements found across many evidence-based practices (Chen et al., 2021; Sutherland et al., 2019). Common practice elements include group contingencies, manipulating the instructional antecedent, problem-solving, social skills instruction, choices, and scaffolding (Chen et al., 2021). Other common elements include teacher feedback, routines, choice,

modeling, self-management, emotional regulation instruction, problem-solving, and building relationships (Sutherland et al., 2019). Educators should include frequent opportunities to respond to maximize engagement. Incorporating peer-mediated interventions had a positive impact on social skills and academic behavior (Wills-Jackson, 2019). This approach allows for flexibility, where educators can rely on their adaptive expertise to use specific elements based on the needs of the students in real time (Chen et al., 2021) and align with the need for individualized interventions (Vanderbilt Kennedy Center, 2014).

Challenges Implementing Strategies

While evidence-based practices (EBPs) for students with EBD exist, implementing them with fidelity has typically not been as successful in the classroom (Criss et al., 2024; Scheibel, 2023; Sutherland et al., 2019). Implementing EBPs requires knowledge, time, and resources to learn and effectively implement (Scheibel, 2023). While teachers may know an effective EBP, they may not have the time, resources, or support to implement it with fidelity (Scheibel, 2023). Teachers self-reported having at least some knowledge of 68% of effective behavioral support practices, at least somewhat knowledgeable of differentiation strategies and classroom management, and aware of approximately 66% of instructional practices effective for students with EBD in inclusive settings (McKenna et al., 2022). Statistically significant differences existed between special educators and general educators. Special educators had more knowledge of behavior practices, differentiation, and classroom management; however, there were no statistically significant differences in knowledge of instructional practices.

Knowledge of practices is important, but educators must implement these practices (McKenna et al., 2022; Scheibel, 2023). Educators reported using 45.45% of behavioral practices, 60% of differentiation strategies, 100% of classroom management practices, and

33.33% of institutional practices at least once a week (McKenna et al., 2022). Statistically significant differences were found between special and general educators, with special educators reporting a higher percentage of using behavioral practices, differentiation, and instructional practices. Overall, all participants reported using less than half of the practices at least once a week, which remains a barrier to inclusive education for students with EBDs.

While EBPs exist, there may not be an EBP that addresses the needs of every individual student (Scheibel, 2023). The existing EBPs are also based on research in specific contexts, which may not apply to every student (Rowe, 2020). While EBPs exist for populations with learning disabilities or autism, the amount for students with complex needs is more limited.

Often, educators will identify non-EBP practices when they cannot identify or feasibly implement an existing EBP (Scheibel, 2023). To mitigate the risk of using methods unsupported by sufficient evidence, educators can use high-leverage practices (HLPs) to adapt non-EBPs to increase effectiveness, with recurrent progress monitoring to identify the potential impact on students.

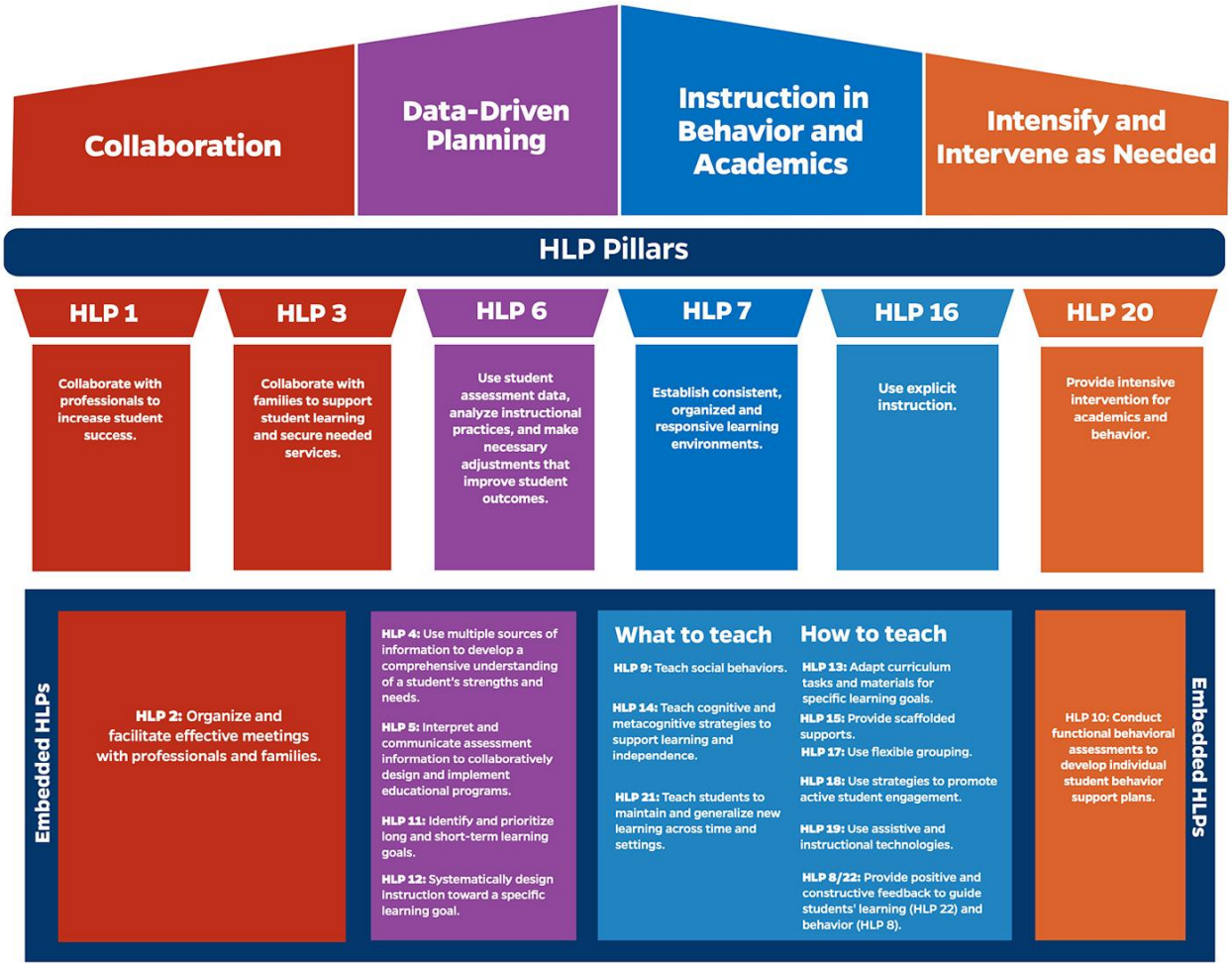
High Leverage Practices for Students with Disabilities

Teachers of students with disabilities have access to different research-based instructional practices; however, it can be challenging to find the time to identify, study, and practice the different research-based strategies (Brownell et al., 2021). In response to this challenge, the Council for Exceptional Children (CEC) and the Collaboration for Effective Educator Development, Accountability, and Reform (CEEDAR) Center identified high-leverage practices (HLPs) for special educators to use with students with disabilities to help improve instruction. Their original publication focused on HLPs in special education (Aceves & Kennedy, 2024). Although HLPs were initially developed for special educators, they were not intended to be

exclusive to them (Council for Exceptional Children, 2024). These practices can and should be used by any educator who works with students with disabilities, including those teaching in general education classrooms. While the practices help students with disabilities, the HLPs are also effective with non-disabled students (Nelson et al., 2022). The newly revised practices are labeled as High Leverage Practices for Students with Disabilities to reflect that HLPs are accessible to all educators who work with students with disabilities (Aceves & Kennedy, 2024). Figure 1 outlines the four domains and twenty-two practices of the HLPs.

Figure 1

High-Leverage Practices



Note. This figure was created by The IRIS Center (2025a, para 2).

The Twenty-Two HLPs

I used the HLPs and the theoretical foundations as the theoretical framework for this study. Twenty-two HLPs help improve instruction and student behavior and are "foundational to effective instruction" (Brownell et al., 2021, p. 13). When successfully applied in classrooms, the practices have improved student outcomes. These revised practices fall under four domains: collaboration, data-driven planning, instruction in behavior and academics, and intensify and intervene as needed (Aceves & Kennedy, 2024). Within each domain, six pillars identify the most valuable practices within these domains, along with embedded practices within the pillars.

Collaboration. The first domain highlights the importance of stakeholder collaboration to identify student strengths and needs and effectively create and monitor individualized education programs (IEPs) to maximize student success (Billingsley et al., 2024). The domain of collaboration includes two pillars and one embedded practice.

Domain One Pillars. "HLP 1: Collaborate with professionals to increase student success" (Billingsley et al., 2024, p. 27). Special educators must collaborate through sharing, problem-solving, and planning with all professionals who work with students with disabilities. These can include general educators, support staff, and paraprofessionals. Collaboration allows the team to meet the student's needs, and supports and services help students progress towards their goals (Billingsley et al., 2024).

"HLP 3: Collaborate with families to support student learning and secure needed services" (Billingsley et al., 2024, p. 27). The families of students with disabilities are vital stakeholders in their children's education. Special educators must work with families to understand the student's strengths and needs and inform families of the resources and processes available to help their child succeed (Billingsley et al., 2024).

Domain One Embedded Practice. "HLP 2: Organize and facilitate effective meetings with professionals and families" (Billingsley et al., 2024, p. 27). Special educators are the case managers and facilitators of all meetings with stakeholders regarding the progress and needs of students with disabilities. They are responsible for effectively organizing and facilitating these meetings to foster effective communication and planning to best meet the needs of students with disabilities (Billingsley et al., 2024).

Data Driven Planning. Teachers must accurately assess their students to provide appropriate support, develop effective plans, and help families understand their children's strengths and needs (Cornelius et al., 2024). This domain has one pillar and four embedded practices.

Domain Two Pillar. The identified pillar for data driven planning is "HLP 6: Use assessment data, analyze instructional practices, and make necessary adjustments that improve student outcomes" (Cornelius et al., 2024, p. 49). Educators need to use various data to create appropriate goals for students with disabilities. Once these goals are developed, it is vital that teachers not only implement effective instructional practices but reflect and analyze the impact on the progress towards those goals. Through analysis, teachers can identify how to adjust instruction best to meet the student's needs (Billingsley et al., 2024).

Domain Two Embedded Practices. The four embedded practices for this domain are:

> HLP 4: Use multiple sources of information to develop a comprehensive understanding of a student's strengths and needs. HLP 5: Interpret and communicate assessment information to collaboratively design and implement educational programs. HLP 11: Identify and prioritize long and short-term learning goals. HLP 12: Systematically design instruction toward a specific learning goal. (Cornelius et al., 2024, p. 49)

These embedded practices support HLP 6 by ensuring teachers are using different types of assessments and information to understand what adjustments may be needed to support student success, possessing the ability to explain the information to stakeholders, and ultimately using this information in collaboration with those individuals to develop appropriate goals and the best instructional practices to use to meet those goals (Cornelius et al., 2024).

Instruction in Behavior and Academics. The highlight of this domain focuses on what teachers are teaching and how they teach, relating to academic and behavioral instruction (Lancaster et al., 2024). This domain contains two pillars and nine embedded practices.

Domain Three Pillars. The two pillars for this domain are: "HLP 7: Establish a consistent, organized, and responsive learning environment. HLP 16: Use explicit instruction" (Lancaster et al., 2024, p. 79). Using explicit instruction is a systematic approach where teachers gradually release "responsibility for learning" (Lancaster et al., 2024, p. 85). This practice is important with students with disabilities because it allows educators to make content accessible to all learners by identifying essential content and scaffolding using "supported and independent practice with feedback" (Lancaster et al., 2024, p. 85) so students can learn and generalize the skills and information taught. HLP 7 guides teachers to provide consistent feedback and ensure students understand expectations and routines (Lancaster et al., 2024).

Domain Three Embedded Practices. Nine embedded practices fall under HLP 7 (Lancaster et al., 2024). The embedded practices focus on what to teach and how to teach. The embedded practices categorized under what to teach are:

HLP 9: Teach Social Behaviors

HLP: 14: Teach cognitive and metacognitive strategies to support learning independence.

HLP 21: Teach students to maintain and generalize new learning across time and settings. (Lancaster et al., 2024, p. 79)

Teachers need to explicitly teach social, emotional, and behavioral skills to students with disabilities (Lancaster et al., 2024). In support of this, teachers should instruct students on strategies that promote "learner self-regulation and independence" (Lancaster et al., 2024, p. 100). This includes problem-solving, regulation, organization, and self-monitoring. Instruction can be explicit or integrated into lessons to generalize skills through authentic learning experiences (Lancaster et al., 2024).

The embedded practices categorized under how to teach are:

HLP 13: Adapt curriculum tasks and materials for specific learning goals.

HLP 15: Provide scaffolded supports.

HLP 17: Use flexible grouping.

HLP 18: Use strategies to promote active student engagement.

HLP 19: Use assistive and instructional technologies.

HLP 8/22: Provide positive and constructive feedback to guide students' learning (HLP 22) and behavior (HLP 8). (Lancaster et al., 2024, p. 79)

Explicit instruction is only one piece of the effective practices puzzle. Educators must also understand that students require individualized adaptations or modifications, whether that is scaffolding the academic content, providing a variety of grouping sizes, identifying which engagement strategies and tools work best, and providing feedback to students on their academic efforts and behavioral actions (Lancaster et al., 2024).

Intensity and Intervene as Needed. This final domain "reinforces the importance of a full continuum of supports and services for students with disabilities" (Hayes et al., 2024, p.142).

Educators require specialized skills to provide more intensive support for students with disabilities effectively. Intensify and Intervene as Needed is targeted for the Tier 3 intervention group, which includes students who have been identified as needing the most intensive supports. While the HLPs are meant for any educator teaching students with disabilities, special educators with the training and expertise most likely use this domain to implement Tier 3 interventions. This domain has one pillar and one embedded practice (Hayes et al., 2024).

Domain 4 Pillar. "HLP 20: Provide Intensive Instruction for Academics and Behavior" (Hayes et al., 2024, p. 142). This HLP highlights the importance of educators providing instruction for individuals or small numbers of students with similar intensive academic or behavioral needs. Progress is monitored and analyzed frequently to ensure the instruction has a positive impact (Hayes et al., 2024).

Domain 4 Embedded Practice. "HLP 10: Conduct Functional Behavioral Assessments to Develop Individualized Student Behavior Support Plans" (Hayes et al., 2024, p. 142). Functional behavioral assessments (FBAs) are individualized behavioral assessments that aggregate multiple forms of data to identify the function of a non-preferred behavior and what is maintaining that behavior. These are used to create a behavior support plan for individual students to help teach them replacement behaviors and/or alter the environment in which the behaviors occur. Frequent data collection and analysis are conducted to ensure the behavior plan has a positive impact (Hayes et al., 2024).

Culturally Inclusive Pedagogies and Practices in HLPs

The updated HLPs are rooted in culturally inclusive pedagogies and practices (CIPP) (Taylor & Monaco, 2024). Educators can more effectively address all students' academic, behavioral, social, and emotional needs by being more culturally aware and inclusive. CIPP

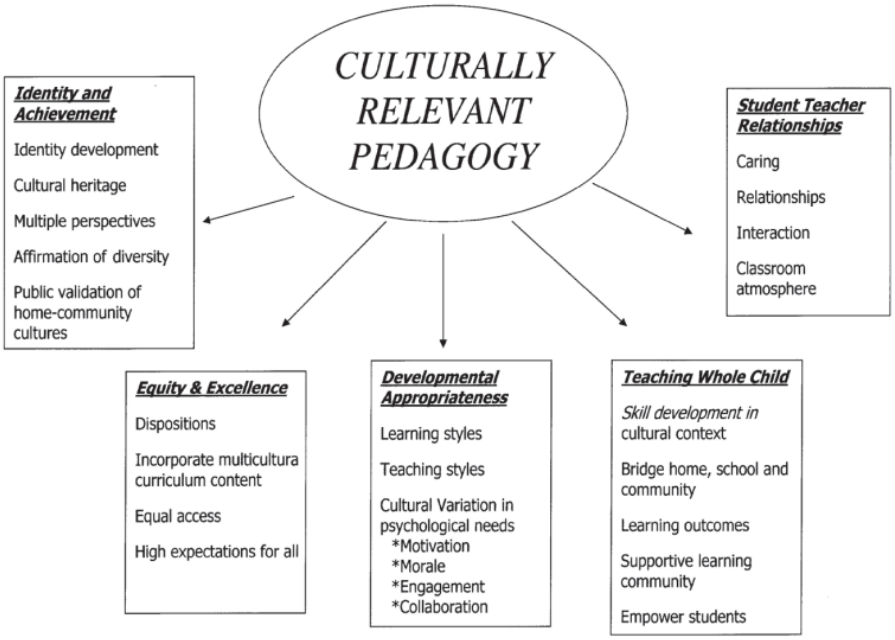
guides educators to see students as "complex individuals" (Taylor & Monaco, 2024, p. 19) with "multilayered identities, experiences, and cultures" (Taylor & Monaco, 2024, p. 19). These guiding ideas help educators to understand and address the complex, individualized needs of students with EBD (Smith et al., 2018; Sutherland et al., 2019) by understanding that the needs of these students are also intertwined with their identities, cultures, and environments (Taylor & Monaco, 2024). These practices are built upon culturally relevant pedagogy, culturally responsive teaching (CRT), and intersectionality.

Culturally Responsive Pedagogy. Gloria Ladson-Billings originated culturally responsive pedagogy to address the need to embed students' social and cultural backgrounds into pedagogy and how it impacts their success (Ladson-Billings, 1995). CRP emphasizes the importance of remaining nonjudgmental of students' cultural backgrounds and intentionally being inclusive of their cultures (Brown-Jeffy & Cooper, 2011). It was further developed by Brown-Jeffy and Cooper (2011) into culturally relevant pedagogy.

Culturally Relevant Pedagogy. The five themes identified in culturally relevant pedagogy are: "identity and achievement, equity and excellence, developmental appropriateness, teaching the whole child, and student-teacher relationships" (Brown-Jeffy & Cooper, 2011, p. 71). Figure 2 outlines the five themes in culturally relevant pedagogy.

Figure 2

The Principles of Culturally Relevant Pedagogy



Note. This figure was created by Brown-Jeffy and Cooper (2011, p. 72).

Identity and Achievement. Identity and achievement focus on "identity development, cultural heritage, multiple perspectives, affirmation of diversity, and public validation of home-community cultures" (Brown-Jeffy & Cooper, 2011, p. 72). Teachers need to be aware of their own identities and how that impacts their perspectives, as well as the identities and perspectives of their students. Becoming aware of these differences is important when intentionally affirming the identities and cultures of others as assets (Brown-Jeffy & Cooper, 2011).

Equity and Excellence. The theme of equity and excellence encompasses the concepts of "dispositions, incorporation of multicultural curriculum content, equal access, and high expectations" (Brown-Jeffy & Cooper, 2011, p. 74). These concepts focus on ensuring teachers give the students what they need, highlighting the importance of differences and differentiation

while holding high expectations for all (Brown-Jeffy & Cooper, 2011). It also includes the importance of ensuring students see themselves in the curriculum in a positive light.

Developmental Appropriateness. This theme focuses on "learning styles, teaching styles, and cultural variation in psychological needs" (Brown-Jeffy & Cooper, 2011, p. 75). Teachers must be aware of their students' developmental needs and stages, provide instruction, and adapt their teaching styles to meet these developmental needs through a cultural lens (Brown-Jeffy & Cooper, 2011). This theme includes understanding that people and cultures have varying psychological needs and motivations.

Teaching the Whole Child. The theme of teaching the whole child outlines the following concepts: "skill development in a cultural context, home-school-community collaboration, learning outcomes, supportive learning community, and empowerment" (Brown-Jeffy & Cooper, 2011, p. 76). Teachers need to look beyond the surface-level needs of students and "remember the needs of the total child" (Brown-Jeffy & Cooper, 2011, p. 76). Teachers need to consider the cultural backgrounds, beliefs, and priorities of families and how these intersect to empower students to be successful (Brown-Jeffy & Cooper, 2011).

Student-Teacher Relationships. The final theme of culturally relevant pedagogy focuses on "caring, relationships, interaction, and classroom atmosphere" (Brown-Jeffy & Cooper, 2011, p. 77). Teachers are a significant part of their students' lives and must be aware of how they cultivate caring relationships with all their students (Brown-Jeffy & Cooper, 2011). They must find ways to connect with their students and create opportunities for them to connect with one another by creating a safe classroom environment that fosters community development.

Culturally Sustaining Pedagogy. Culturally sustaining pedagogy was developed to further the importance of valuing multiethnic and multilingual environments (Paris, 2012).

Pedagogy needs to be responsive, relevant, and supportive of students' cultural and linguistic differences.

Intersectionality. Intersectionality focuses on the intersection of multi-level identities and how these identities shape individuals' experiences, including their privileges and experiences with oppression (Garcia, 2024). Intersectionality challenges educators "to consider the layers of oppression that affect students across these diversities" (Taylor & Monaco, 2024, p. 21) as they develop their instructional practices and integrate culturally inclusive pedagogies and practices (Taylor & Monaco, 2024).

Culturally Inclusive Pedagogies and Practices and High Leverage Practices. CIPP within the HLPs highlights culturally responsive pedagogy and teaching, culturally relevant pedagogy and teaching, and culturally sustaining pedagogy and teaching. Culturally responsive teaching directs teachers to take a cultural lens with curriculum and teaching strategies to make them more meaningful to their students (Gay, 2018). Culturally relevant teaching guides teachers to be aware of the cultural identities of their students, supporting those identities alongside academic support (Ladson-Billings, 1995). Culturally sustaining teaching emphasizes the importance of supporting cultures in the classroom through recognizing and sharing cultural aspects throughout the educational process (Taylor & Monaco, 2024). CIPP and intersectionality guide perspectives to be considered throughout all the HLPs to ensure educators understand and address students with disabilities and their families in an inclusive, culturally appropriate manner. Teachers can use these guiding frameworks alongside HLPs to reflect on their cultures and identities, recognizing how they may differ from their students. This reflection can help educators connect with their students and communities. It is essential for teachers to understand their students' diverse identities and backgrounds and ensure they are teaching from an asset-

based perspective that includes all students' identities and cultures (Brown-Jeffy & Cooper, 2011; Paris, 2012; Taylor & Monaco, 2024). HLPs with a culturally inclusive lens provide a solid foundation for effective teaching when applied to instructional approaches. (Taylor & Monaco, 2024). The HLPs and foundational frameworks in CIPP were the theoretical frameworks that guided this study.

Project-Based Learning

Project-based learning is part of a larger category of teaching methodology called inquiry-based learning (Savery, 2006). Inquiry-based learning includes project-based learning, problem-based learning, and design-based learning. Inquiry-based learning focuses on answering questions through student research and problem-solving. Problem-based learning focuses on having students set goals for solving a problem and is more autonomous. In contrast, project-based learning allows teachers to scaffold, model, and guide students. The terminology can be confusing because project-based learning and problem-based learning share the same acronym: PBL. To avoid confusion, I will use PjBL for project-based learning, following the convention established by other researchers (Culclasure et al., 2019; Mulcahy & Wertz, 2021).

Project-based learning (PjBL) is a cooperative teaching method that involves learning by doing (Fischer, 2023). More than a culminating project, the students' process is a key aspect of PjBL. PjBL traces its roots back to John Dewey, who believed children should be active learners in collaborative, experiential learning environments where the teacher was a facilitator rather than a lecturer (Virani-Murji, 2020). Project-based learning (PjBL) is currently defined as "a teaching method in which students gain knowledge and skills by working for an extended period to investigate and respond to an authentic, engaging, and complex question, problem, or challenge" (PBLWorks, 2024b, para. 1).

High Quality Project-Based Learning

PBLWorks is an organization under the Buck Institute for Education that focuses on research, development, and support for teachers and schools using project-based learning (PBLWorks, 2024a). They developed two gold-standard guides for PjBL: the "Seven Essential Project Design Elements" (PBLWorks, 2024e) and "Seven Project Based Teaching Practices" (PBLWorks, 2024e). The design elements focus on developing quality PjBL units, and the practices focus on assessing and improving existing PjBL. Both the design elements and teaching practices focus on the goals of "key knowledge, understanding, and success skills" (PBLWorks, 2024e). In addition to these guides, the Buck Institute for Education developed, in partnership with teachers and researchers, the High Quality Project-Based Learning Framework (HQPBL), which outlines six criteria that must be present for a PjBL experience to be considered high quality (The Buck Institute for Education, 2018).

Seven Essential Project Design Elements. The essential project design elements model assists professionals in developing quality PjBL units for their classes (PBLWorks, 2024e). The seven elements are a challenging problem or question, sustained inquiry, authenticity, student choice and voice, reflection, critique and revision, and public product (PBLWorks, 2024b). A PjBL unit must begin with a problem or question that provides an "appropriate level of challenge" (PBLWorks, 2024b, para. 3). In efforts to solve this problem or question, students must be engaged in sustained inquiry where they are "posing questions, finding resources, and applying information" (PBLWorks, 2024b, para. 4). It is important that the unit focuses on a real-world situation or context that is relevant to the students and that they can make decisions throughout the project. Both teachers and students engage in reflective practices while critiquing

each other's work and revising along the way. The culmination of a unit results in a public product that the students share with people outside their class.

Seven Project-Based Teaching Practices. Project-based teaching practices provide a framework for educators to use when assessing and improving their practice (PBLWorks, 2024e). The seven practices are design and plan, align to standards, build the culture, manage activities, scaffold student learning, assess student learning, and engage and coach (PBLWorks, 2024c). Teachers begin planning their unit by identifying their context and how it relates to their students while ensuring the academic standards are addressed. Building culture is a vital element of PjBL. Teachers need to create a culture that promotes "student independence and growth, open-ended inquiry, team spirit, and attention to quality" (PBLWorks, 2024c, para. 5). Throughout the project, teachers must ensure they manage activities as a guide and coach while scaffolding for learners with diverse needs. Formative and summative assessment opportunities are used for individuals, peer assessments, and the group.

High Quality Project-Based Learning Framework. Educators and researchers developed the HQPBL framework to determine what students would experience when engaging in high-quality PjBL (The Buck Institute for Education, 2018). This framework can be used by educators when designing and teaching PjBL experiences. The framework outlines six criteria to use as a starting point for HQPBL: intellectual challenge and accomplishment, authenticity, public product, collaboration, project management, and reflection (The Buck Institute for Education, 2018). Students should be engaging in projects requiring them to think critically about complex issues and take time to complete them. These projects must be relevant and meaningful to their lives, with student voice and choice embedded in the work. Projects are meant to be publicly shared so students receive ongoing feedback from their peers, teachers, and

others outside the classroom. While some aspects of the learning experience can be completed individually, collaboration on project work must also be included in HQPBL to give students real-world experiences working on a team. Students must engage in project management within their team, whether managing their time, tasks, resources, or design thinking. HQPBL experiences prompt students to reflect on their work through self-assessment and consider ways to improve. These six criteria are "an essential baseline, but they are not all-encompassing" (The Buck Institute for Education, 2018, p. 2). Classroom culture, student voice, and "the fundamental belief that all students can learn" (The Buck Institute for Education, 2018, p. 2) are valuable in implementing PjBL (The Buck Institute for Education, 2018).

PjBL and Students Without Disabilities

The effects of PjBL on students without disabilities are well documented, with positive outcomes related to academic outcomes (Aydin & Mutlu, 2023; Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Gomez-Pablos et al., 2017; Krajcik et al., 2023; Revelle et al., 2020; Saavedra et al., 2021; Schneider et al., 2022; Sormunen et al., 2019; Suryanti et al., 2024; Wang et al., 2022; Zhang et al., 2023), behavior, and social-emotional learning (Culclasure et al., 2019; Febriana, 2017; Krajcik et al., 2023; Juniar et al., 2023; Lee et al., 2015; Samsudin et al., 2020; Wang et al., 2022). Students reported positive feelings towards their PjBL experiences, including helping them learn, excitement about their learning, and being able to work with others (Culclasure et al., 2019; Mariadas & Yusop, 2024). Teachers also reported positive opinions, with improvement in student achievement, enthusiasm, and motivation compared to other teaching approaches (Culclasure et al., 2019; Zhang et al., 2023).

Several studies show statistically significant differences in favor of PjBL (Aydin &Mutlu, 2023; Duke et al., 2021; Suryanti et al., 2024; Zhang et al., 2023). Elementary students had

statistically significant growth ($p=.05$) in scientific literacy skills (Suryanti et al., 2024), social studies knowledge ($p<.001$) (Duke et al., 2021), academic scores ($p=.005$) (Aydin & Mutlo, 2023), and reading skills ($p<.05$) (Zhang et al., 2023). There were mixed results on motivation, with one study finding no impact (Duke et al., 2021) and another stating that it "intrinsically motivates students who learn history" (Marias & Yusop, 2024, p. 72). One study earned a Tier Two moderate rating on What Works Clearinghouse (What Works Clearinghouse, 2024). This rating was developed as evidence tiers through the Every Student Succeeds Act (ESSA) for schools to identify what products, policies, programs, or practices work (What Works Clearinghouse, n.d.). A Tier Two rating on What Works Clearinghouse means the targeted item has moderate evidence of effectiveness. This rating is attained by meeting the standards with or without reservations, having statistically significant positive results, at least 350 students, and at least two educational sites in the study. Researchers conducted a randomized control trial with 4,000 students in high school physics and chemistry. They found that students in the experimental group performed an average of 0.2 standard deviations above the control group on a summative science assessment, with $p < 0.05$ (Schneider et al., 2022).

PjBL and Students with Disabilities

While the benefits of PjBL have been documented for students without disabilities, research on PjBL with students with disabilities is limited (Boardman & Hovland, 2024; Wertz & Mulcahy, 2024). Studies on students with disabilities focused on those with speech, reading, and writing disabilities (Berkeley et al., 2019), learning disabilities (Lu et al., 2022; Sormunen et al., 2019), students who were blind, deaf, or intellectually disabled (Eldiva & Azizah, 2019), or in mixed settings (Purnomo et al., 2022; Hovey & Ferguson, 2014). Mixed results included constraints regarding self-efficacy (Berkeley et al., 2019), positive impact on creativity and

learning outcomes (Lu et al., 2022), improved critical thinking skills (Eldiva & Azizah, 2019), and the need for more intensive support (Sormunen et al., 2019).

Students with disabilities report that PjBL positively impacted their collaboration skills, ability to have choices in their learning, and the authenticity of the experience (Boardman & Hovland, 2024). They also reported needing more time throughout the process, navigating the stresses of group work, such as when a team member does not contribute, and needing more support from their teachers.

Research that included teacher perspectives established that educators view PjBL as an effective approach for students with disabilities (Hovey & Ferguson, 2014; Purnomo et al., 2022). Teachers believed that PjBL improved student enthusiasm, increased communication and social skills, and gave them more control over their behaviors (Purnomo et al., 2022). The current findings demonstrate the possibility that PjBL may be an effective method when adapted for students with disabilities (Boardman & Hovland, 2024; Eldiva & Azizah, 2019; Hovey & Ferguson, 2014; Lu et al., 2022; Purnomo et al., 2022; Sormunen et al., 2019).

PjBL and EBD Research

Initial findings on PjBL with students with EBD demonstrate the possibility of a positive impact on this student population (Foster & Shriner, 2023; Taylor et al., 2018; Taylor et al., 2024). One study suggests using PjBL in an art context (Taylor et al., 2018), while another found positive impacts on the students and teachers (Taylor et al., 2024). Art-based PjBL can allow for more opportunities for student expression outside typical academic contexts, allowing for a more strengths-based approach that also "helps develop positive relationships that students with challenging behaviors need" (Taylor et al., 2018, p. 64). PjBL improved student behavior related to prosocial behavior and task completion (Taylor et al., 2024). It also improved job satisfaction

scores for teachers and paraprofessionals using PjBL with students with EBD. Teachers in one residential special education school for students with significant emotional and behavioral challenges noted that negative behaviors decreased when students were engaged in PjBL lessons, with an increase in engagement and academics (Foster & Shriner, 2023). While limited, the initial research is promising.

HLP and PjBL

While PjBL can improve engagement and provide an authentic learning experience for students, educators must address the barriers to students with disabilities (IRIS Center, 2024). Barriers can include executive functioning needs and the ability to generalize information and skills. Though explicit instruction is a HLP student with disabilities benefit from, this does not preclude them from learning successfully in a PjBL setting (Boardman & Hovland, 2024). Students with disabilities need support to be successful in an inquiry-based environment (Rizzo & Taylor, 2016).

PjBL naturally incorporates many HLPs that are effective for students with disabilities (Mulcahy & Wertz, 2021). There are 11 similarities between the twenty-two HLPs and elements of PjBL (Wertz & Mulcahy, 2024). These similarities include:

- PjBL's use of open-ended questioning and various artifacts reflects its use of multiple sources of information to understand students' strengths and needs (HLP 4).
- Using assessment data to adjust instruction (HLP 6) aligns with the emphasis on continuous feedback in PjBL.
- Creating an environment with expected norms and routines with student collaboration aligns with HLP 7 by constructing a "consistent, organized, and respectful learning environment" (Wertz & Mulcahy, 2024, p. 4).

- PjBL incorporates collaborative learning opportunities, which align with HLP 9.
- Designing instruction towards a specific learning goal (HLP 12) and incorporating assistive and instructional technology (HLP 19) is reflected in teachers developing units with targeted learning goals for students.
- When adapting instruction and materials for students through scaffolding and explicit instruction, teachers utilize HLP 13 and 15 (Wertz & Mulcahy, 2024).
- PjBL allows for flexible grouping (Mulchay & Wertz, 2020) and aligns with HLP 17: "use flexible grouping" (Aceves & Kennedy, 2024, p. 15).
- Promoting active student engagement (HLP 18) through strategies used throughout a PjBL unit (Wertz & Mulcahy, 2024).
- PjBL embeds opportunities for peer and teacher feedback (Mulcahy & Wertz, 2021), aligning with HLP 22.
- Monitoring student progress and providing support based on individual student needs (Mulcahy & Wertz, 2021; Wertz & Mulcahy, 2024) aligns with data-driven planning (Cornelius et al., 2024) and instruction in behavior in academics (Aceves & Kennedy, 2024).

While current research integrating HLPs for students with disabilities in PjBL is limited, the similarities are promising (Boardman & Hovland, 2024; Mulcahy & Wertz, 2021).

Gaps in the Literature and Further Research

Though PBLWorks' vision is to have project-based learning for all (PBLWorks, 2024d), the research on PjBL with students with disabilities is limited (Boardman & Hovland, 2024; Wertz & Mulcahy, 2024). This scarcity of research is even more pronounced for students with EBD (Foster & Shriner, 2023; Taylor et al., 2024). Given the positive impact it has had on

students with disabilities (Boardman & Hovland, 2024; Eldiva & Azizah, 2019; Foster & Shriner, 2023; Hovey & Ferguson, 2014; Lu et al., 2022; Purnomo et al., 2022; Taylor et al., 2024), and without disabilities (Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Gomez-Pablos et al., 2017; Revelle et al., 2020; Samsudin et al., 2020; Sormunen et al., 2019), as well as the features it shares with many of the HLPs (Wertz & Mulcahy, 2024), the need for more research with this population is necessary. PjBL may present challenges in implementation, such as scheduling, group dynamics, and working with students who are frequently absent or not in the classroom (Mulcahy & Wertz, 2021). Examining teachers' implementation of PjBL with students with EBD and their perspectives on its impact can offer valuable insights. This information can guide educators to effectively use PjBL to address the complex needs of students with EBD (Wertz & Mulcahy, 2024).

Chapter Summary

Students with EBD are at an elevated risk for adverse outcomes (Chen et al., 2021; Gersib & Mason, 2023; National Center for Education Statistics, 2024; Office of Special Education Programs, 2020). The limited number of effective strategies (Chen et al., 2021) is challenging to implement (Gilmour et al., 2022; Smith et al., 2018; Sutherland et al., 2019). The need for flexibility for their complex needs (Smith et al., 2018) suggests using common strategies across interventions in adaptive ways (Chen et al., 2021) and incorporating HLPs into non-EBPs (Aceves & Kennedy, 2024). PjBL has been successful with non-disabled students (Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Gomez-Pablos et al., 2017; Revelle et al., 2020; Samsudin et al., 2020; Sormunen et al., 2019) and shares similarities with HLPs (Wertz & Mulcahy, 2024). The limited research conducted with students

with EBD demonstrates the need for more research on how PjBL is used with students with EBD and its impact (Taylor et al., 2024).

Chapter III. Methodology

This collective case study aimed to understand teachers' experiences and perceptions using project-based learning (PjBL) with students with emotional and behavioral disorders (EBD). Students with EBD experience challenges that result in adverse outcomes both in school (Chen et al., 2021; Didion et al., 2018; Kumm et al., 2021; Enis et al., 2018; Hutchins et al., 2020; Mathews et al., 2021; National Center for Education Statistics, 2024; Smith et al., 2018) and out of school (Chen et al., 2021; Lipscomb et al., 2017; Offerman et al., 2022). While evidence-based practices exist, implementing them with fidelity has not been as successful in the classroom (Criss et al., 2024; Scheibel, 2023; Sutherland et al., 2019). Educators can use high-leverage practices in conjunction with instructional methods that may not be evidence-based to meet the needs of these students. Project-based learning (PjBL) has been successful in environments with non-disabled students (Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Gomez-Pablos et al., 2017; Revelle et al., 2020; Samsudin et al., 2020; Sormunen et al., 2019) and shares similarities with HLPs (Wertz & Mulcahy, 2024). The limited research conducted with students with EBD demonstrates the need for more research on how PjBL is used with students with EBD and its impact (Taylor et al., 2024). This qualitative case study will collect data to understand the experience and perspectives of teachers who use PjBL with students with EBD.

Rationale for Approach

Case study research is a qualitative approach that investigates "a real-life, bounded system (case) or multiple bounded systems (cases) over time, through detailed, in-depth data collection involving multiple sources of information" (Creswell & Poth, 2018, pp. 96-97). Qualitative research is used when a problem has variables that are difficult to measure (Creswell

& Poth, 2018). A qualitative case study allowed me to explore the complex challenges associated with teaching students with EBD through the story of teachers' experiences, focusing on research conducted in a real-life context (Creswell & Poth, 2018). Data collection included interviews and artifacts, allowing me to understand what the participants experienced, how they implemented PjBL, how teachers felt PjBL impacted their students, and their perceptions of how successful they felt using PjBL as an instructional methodology.

Qualitative researchers must examine participants' perspectives in their natural environment to understand a phenomenon (Suter, 2012), allowing researchers to discover what meaning people attribute to a problem (Creswell & Poth, 2018). Researchers for case studies use "real life cases that are in progress so that they can gather accurate information not lost by time" (Creswell & Poth, 2018, p. 97). This design allowed me to gather relevant and timely data (Creswell & Poth, 2018) on educators currently using PjBL with students with EBD.

One issue is chosen in a collective case study, and the researcher identifies multiple cases to research (Creswell & Poth, 2018). Multiple cases highlight different perspectives, and while generalizing qualitative research is difficult, researchers can use multiple cases to mitigate this issue. A collective case study allowed me to gain perspectives from three different teachers. This design offered the opportunity to understand teachers' insights regarding the perceived effectiveness and feasibility of PjBL with students with EBD. This approach contributed valuable insights into the effectiveness of PjBL for students with EBD.

Research Method and Design

This qualitative, collective case study examined teachers' experiences and perceptions using PjBL with students with EBD. The research question was: What are the experiences and

perceptions of teachers who use project-based learning with students with emotional and behavioral disorders?

Site and Sample Selection

Criterion sampling is used in qualitative research to assist in identifying participants with similar characteristics and experiences (Creswell & Poth, 2018). This sampling method is helpful in qualitative research because it can "reflect the purpose of the study and guide in the identification of information-rich cases" (Merriam & Tisdell, 2016, p. 97). Using criterion sampling for this study, I identified three participants with similar characteristics who all experienced using PjBL with students with EBD in grades 6-12.

The criteria for the participants for this study were the following:

1. Participants must have at least one year of teaching experience with students with EBD.
2. Participants must have taught a complete project-based learning unit in a class with students with EBD within the prior school year.

Students with EBD may be identified under different primary disability categories in their IEP, such as ED, other health impairment (OHI), intellectual disabilities (ID), or autism. For this study, I focused specifically on teachers with students whose primary disability category was ED or a combination of ED and OHI to minimize potential data inconsistencies and maintain a clear focus on a defined group. Teachers were asked to reflect only on their experiences with students identified under these categories, rather than those with other classifications, such as autism. The term EBD is used throughout this report to refer to students who were identified with ED or ED and OHI in school settings, received emotional support services, had a positive behavior support

plan included in their IEP, or whose IEP includes services and supports related to social or emotional skill development or functional behavior (22 PA Code Chapter 14 § 14.131, para. iv). EBD is also a more inclusive and less stigmatizing way to describe these students and is used by other researchers in the field (Culclasure et al., 2019; Kumm et al., 2021).

Requiring participants to have at least one year of teaching experience with students with EBD ensured they had a minimum amount of time to draw upon when relating their experiences. Requiring participants to have completed at least one unit of study where PjBL was the instructional approach in a classroom with students with EBD was also essential to maintaining relevant and timely data (Creswell & Poth, 2018). Requiring participants to have completed an entire unit ensured their experiences were similar. This assisted in data analysis when identifying "common experiences, themes, and the overall essence of the experience" (Creswell & Poth, 2018, p. 152). The addition of requiring the unit to be completed in the past school year could have also increased the likelihood of participants more fully recalling their experiences.

Participant Access

I was able to gain three participants through three different methods. I emailed the network of STEM points of contact across Pennsylvania (see Appendix A), requesting that they forward information to teachers or individuals they believe are using PjBL with students with EBD. I gained one participant through this method and emailed them (see Appendix B) with an initial invitation to participate and a brief Google form (Appendix C).

The second method was through social media invitations to participate (see Appendix D). I posted on my professional LinkedIn profile, inviting interested people to click a link directing them to the initial Google form (See Appendix C). I had one person complete the form indicating their interest. Following these methods, I contacted several of my professional colleagues who

had authorization to provide access to other educators across Pennsylvania. They shared the information, and I obtained an additional two participants.

Once participants completed the initial Google form (see Appendix C) indicating their interest, I analyzed it to ensure they met the study's criteria. Three out of the four people who indicated interest met the criteria for the study. I emailed these participants individually to schedule the initial meeting (see Appendix E) to review the informed consent (Appendix F). Upon receiving informed consent, follow-up meetings were scheduled to interview each participant individually.

Participant Descriptions. The three participants in this study are referred to under the pseudonyms of Mrs. G, Mr. Sparks, and S.L. At the time of this study, Mrs. G. had been in education for 25 years, including time as a paraprofessional. She instructed students with disabilities in different settings, including self-contained and supportive areas. During this study, she was a teacher at a military academy in Pennsylvania, supporting students with and without disabilities. At the time of this study, Mr. Sparks was an eighth-grade STEM teacher at a public school district in Pennsylvania. He taught for ten years, mainly in math, with the 2024-25 school year being his first full year as a STEM teacher. At the time of this study, S.L. was a consultant at an intermediate unit in Pennsylvania; however, she left the classroom in December 2024, where she was an emotional support teacher in a self-contained classroom. All three participants have experience teaching students with emotional and behavioral disorders in the 2024-2025 school year using project-based learning methods.

Data Collection Procedures

Triangulation of multiple data sources is used to evaluate whether themes and evidence appear across different areas (Creswell & Poth, 2018; Suter, 2012). Multiple data sources in case

study research include observations, interviews, documents, and audiovisual materials (Creswell & Poth, 2018). This study included participant interviews and documents the participants provided, including lesson plans, teacher notes, and student artifacts related to the PjBL units.

Interviews were conducted through a personal account using Zoom's video call feature to protect the participants' privacy and ensure that work accounts were not associated with recorded interviews. A private, password-protected link was sent to the participants' personal emails. I used Zoom's lock meeting feature to lock the call once it begins, preventing anyone from logging in after they enter the room (Zoom, 2024). A unique password-protected link with a lock meeting feature will be used to protect the participants' privacy.

The interview was recorded using Zoom's built-in recording feature for data collection. This recording was saved on a password-protected account through Zoom's cloud feature. File names will be number-coded, leaving all personally identifiable information off naming conventions. Transcriptions were created through Zoom and edited for accuracy.

Interview Protocol. Forty-eight hours before the interview, I emailed the participant a copy of the questions (see Appendix G). In one instance, due to scheduling constraints, I gave the participant the interview questions twenty-four hours before the interview. As suggested by Brinkmann and Kvale (2015), before the recording began, I reminded participants of the purpose of the study, that the interview would be recorded, and that their information would be kept protected and anonymous. I also asked them if they had any questions before I began a semi-structured interview (see Appendix H).

Open-ended questions and follow-up probes were developed (see Appendix H) by considering thematic research questions (Moustakas, 1994) with high-leverage practices (HLPs) and culturally inclusive pedagogy and practices (CIPPs) guides. Themes in questioning included

adaptations and modifications to PjBL, inclusivity, advantages or positive outcomes of using PjBL with students with EBD, and challenges or obstacles faced. Themes such as advantages and challenges have also been used in other research regarding teacher perspectives and PjBL (Yang et al., 2021). Additional questions related to these themes were asked for clarification, which allowed them to describe their experiences in more detail. After the interview, participants were thanked for their time, and I said I would contact them if I had further questions or clarifications.

Data Management

Interviews were conducted through the video call feature of Zoom on a personal account to protect the privacy of the individuals and ensure that work accounts were not associated with recorded interviews. An individually unique, private, password-protected link was sent to each participant's email. Zoom has a lock meeting feature that allows users to lock the meeting once it has begun, preventing anyone from logging in after the host has locked it (Zoom, 2024). A unique password-protected link with a lock meeting feature protected the privacy of the interviewed individuals.

Each interview was recorded using Zoom's built-in recording feature for data collection. These recordings were saved on a password-protected account through Zoom's cloud feature. File names were number-coded, leaving all personally identifiable information off naming conventions. Transcriptions were made through Zoom. Accuracy checks were performed by reading through the transcripts while listening to the recorded interviews, and necessary edits were made to the written transcripts based on the audio recordings. File names for all artifacts, teacher documents, and researcher notes were number-coded for each participant.

Ethical Considerations

There may be ethical concerns with the perception that I will side with the participants. I reflected on my personal bias towards educators working with students with EBD. I sought three participants to gain multiple perspectives (Creswell & Poth, 2018). I intentionally reflected before interviewing and throughout data analysis to put my feelings aside and conduct an honest report of the data, not just positive results. This included engaging in the epoche process, where qualitative researchers "set aside predilections, prejudices, predispositions, and allowing things, events, and people to enter anew into consciousness" (Moustakas, 1994, p. 84).

The participants' privacy was an additional concern (Brinkmann & Kvale, 2015). I ensured that I gained informed consent and took the time to explain the steps I would take to safeguard their privacy. Data was also saved in password-protected files, with numerical identification to remove all personally identifiable information. I also asked participants to choose a pseudonym that I used for the data report.

In order to prepare for the research process, I completed the CITI Human Research Subjects Training (see Appendix I). I also received approval from the Wilkes University Institutional Review Board before conducting the research.

Data Analysis

Data files did not contain personally identifiable information in their names or within the documents. Interview recordings and analysis files from each data point and analysis iteration were saved with a name and number code. All data analysis files were stored on a password-protected computer via ATLAS.ti 24 and a password-protected Google account provided by Wilkes University.

Before each analysis iteration, I wrote and read a reflection on my biases and perspectives related to the topic to set aside my perceptions (Moustakas, 1994). I also framed the analysis using my research question and theoretical framework. I listened to the interviews while reading the transcripts to identify errors and made the necessary corrections. I used jotting, memoing, and inductive and deductive coding in my analysis process (Miles et al., 2020) with ATLAS.ti 24.

In the initial coding round, I identified words and phrases that reflected the participants' experiences and perceptions of using PjBL with students with EBD. The first three rounds of coding involved analyzing each transcript separately. I started with the first two participants' interviews because there was a time gap between them and the last one. Memoing was completed after each round of data analysis. I reviewed these two transcripts again, beginning with 122 codes between them. After this second round, I grouped similar codes and identified seven themes. Once I finished analyzing the last interview, I conducted another coding round, using existing codes and noting any new information.

After these initial three rounds, I reviewed my theoretical framework and each transcript a fourth time, categorizing and coding data related to my framework, high-leverage practices, and culturally inclusive pedagogy. Following this fourth review, I organized the data into five themes with fifteen coding labels. I then completed two rounds of coding artifacts using these codes. One participant provided six student artifacts and three modified lesson plans, another shared one student artifact, and a third shared two lesson plans. I used a password-protected Google Sheets document to compare codes across cases and identify elements aligned with the theoretical framework. After analyzing all the data, I refined the themes into three overarching themes with nine subthemes.

After the initial analysis, I shared the identified themes with the participants (see Appendix J) and conducted a member check to verify the validity and reliability of my findings (Creswell & Poth, 2018; Merriam & Tisdell, 2016). I also submitted my findings to a peer reviewer (see Appendix K) for an external review (Creswell & Poth, 2018; Merriam & Tisdell, 2016) and asked them to assess the accuracy of my results.

Role of the Researcher

As a former special educator who worked with students with EBD and used methods similar to PjBL, it is my responsibility as a researcher to disclose my biases and experiences to raise awareness of potential researcher bias. I observed many positive outcomes when using PjBL with my students with EBD. My students were more engaged, and there were fewer behavioral issues. While it was more challenging to formally assess their knowledge through these learning experiences, I consider PjBL an effective method for this population. Strategies to reduce bias in the data analysis process were described above.

Validity and Reliability

Several strategies for validity and reliability were used to ensure the study's credibility. First, I disclosed my "bias, values, and experiences" (Creswell & Poth, 2018, p. 261), which helped establish credibility from the researcher's perspective. I also conducted member checking and gathered participant feedback to address validity from their viewpoint. Member checking also contributed to reliability (Merriam & Tisdell, 2016), as participants reviewed the initial findings to verify accuracy (Moustakas, 1994). To enhance validity from the reviewer's perspective, I asked a peer familiar with project-based learning to review the data as an external check (Creswell & Poth, 2018).

After collecting and analyzing all the data, I submitted the initial draft of the findings to the participants for a member check (Creswell & Poth, 2018; Merriam & Tisdell, 2016). Participants received individual emails with a draft of the findings and were asked to respond with comments on the accuracy of the statements (see Appendix J). They had one week to review this before submitting their feedback via email. Any additional clarifying information was included in the final report.

I submitted the findings to a peer reviewer for an external check (Creswell & Poth, 2018; Merriam & Tisdell, 2016). This individual was an educator with experience in project-based learning at the elementary level. They were also a doctoral student studying qualitative research. They were contacted via email and given instructions on what they would be reviewing (see Appendix K). They were asked to review the report and provide constructive feedback within one week. In my final report, I included a section that disclosed my biases and experiences so the reader is fully aware of the potential for researcher bias.

Additional reliability strategies included verifying the accuracy of the transcripts (Creswell & Poth, 2018) and maintaining a detailed audit trail (Merriam & Tisdell, 2016). Besides using computer-generated transcripts for interviews, I reviewed these transcripts for accuracy before starting data analysis (Creswell & Poth, 2018). Merriam and Tisdell (2016) suggest that researchers keep a journal to document each step in the research process. Throughout the study, I examined the procedures and recorded each step to ensure I followed through and reflected.

Summary

A collective case study was conducted to understand teachers' experiences and perspectives using PjBL with students with EBD. Data was collected through interviews and

artifacts from three participants. Analysis was completed through an iterative process, identifying codes and themes. Strategies for validity and reliability included member checking, peer review of data, and deliberate reflection on my biases and experiences at each step of the process.

Chapter IV. Findings

This collective case study sought to understand the experiences and perceptions of teachers using project-based learning (PjBL) with students with emotional and behavioral disorders (EBD). The challenges associated with EBD contribute to adverse outcomes for students in (Chen et al., 2021; Didion et al., 2018; Kumm et al., 2021; Enis et al., 2018; Hutchins et al., 2020; Mathews et al., 2021; National Center for Education Statistics, 2024; Smith et al., 2018) and out of school (Chen et al., 2021; Lipscomb et al., 2017; Offerman et al., 2022). Though evidence-based practices exist, educators struggle to implement them with fidelity (Criss et al., 2024; Scheibel, 2023; Sutherland et al., 2019). High-leverage practices (HLPs) are strategies educators can implement with other instructional methods to assist in meeting the needs of their students. PjBL is an instructional method that shares similarities with HLPs (Wertz & Mulcahy, 2024) and has been successful with non-disabled students (Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Gomez-Pablos et al., 2017; Revelle et al., 2020; Samsudin et al., 2020; Sormunen et al., 2019).

The research question guiding this study was: What are the experiences and perspectives of teachers using PjBL with students with EBD? Participants were selected via professional networks through a criterion sampling method. HLPs and the theoretical foundations of culturally responsive pedagogy and practices served as frameworks for data analysis. This chapter presents the findings from data collected through participant interviews and artifact analysis.

Participant Profiles

Three participants took part in this study. Each one was interviewed and provided artifacts related to their experiences with PjBL and students with EBD. They each chose a pseudonym to protect their privacy.

M.G.

At the time of this study, M.G. was a special education teacher at a military academy in Pennsylvania. In her role, she collaborated with teachers to support students with GED and credit recovery. She also participated in teams focused on transition planning for students preparing for post-secondary education, employment, or independent living. She has worked in education for 27 years as a para-educator and teacher. When this study was conducted, she had been teaching at the military academy for three years.

M.G. holds a bachelor's degree in early childhood education and a Master's degree in adapted physical activity. She is certified in special education K-21 (the equivalent of a Pk-12 special education certificate) and reading and is pursuing certifications in English as a second language and trauma-informed education. Her professional experience spans inclusive and self-contained settings across public school districts, charter schools, intermediate units, and educational agencies. She has worked with students with a wide range of disabilities, including learning disabilities, emotional and behavioral disorders, autism spectrum disorders, intellectual disabilities, and multiple motor impairments. M.G. stated that most of her experience was with students with emotional and behavioral disorders.

B.Q.

At the time of this study, B.Q. was an eighth-grade STEM teacher in a school district in central Pennsylvania. He has a degree in mathematics, an associate's degree in mechanical

technology/engineering, is a certified teacher, and has a M.S. in Instructional Design and Technology. He began his career at a therapeutic school for students with mental health challenges, where he collaborated with other professionals and therapists to meet students' needs. B.Q. explained that this school was a therapeutic school for students who had “more depression, anxiety-based” disorders. Before working in Pennsylvania, he taught as a high school math teacher at a Catholic school in Maryland. He spent 10 years as a math teacher before transitioning to the eighth-grade STEM position in the 24-25 school year. In addition to his role as the STEM teacher, he also tutored students in math. His experiences teaching students with EBD were in inclusive, general education settings.

S.F.

At the time of this study, S.F. was a STEM Innovation Specialist at an intermediate unit in Pennsylvania. She had recently worked as a special education teacher at an intermediate unit and left that position to take on her specialist role in December 2024. The experiences she shared were from her time as a special educator.

S.F. holds a master’s degree in educational/instructional technology and special education, PK-12, and elementary education certifications. She began her educational career as a paraprofessional and has twelve years of teaching experience. Her teaching career was mainly in emotional support classrooms serving students in grades 3-12 and partial hospitalization programs. She primarily worked with students diagnosed with emotional disturbance, but also has experience working with students with autism and specific learning disabilities. S.F. has taught in various settings, including self-contained classrooms and alternative education schools.

Setting Context for Findings

This section offers background information to provide context about the participants' experiences. Each participant shared their understanding and use of PjBL. Although their lessons varied in content, they shared many common features.

Each participant planned lessons grounded in real-world scenarios or problems, incorporating hands-on activities with group collaboration. They designed their own PjBL activities, lessons, and units, some of which integrated various content areas, vocabulary, and research skills. Interview and artifact data also revealed student goals for real-world skills included as learning focuses. All participants provided examples of diverse project-based assessments completed in groups. M.G.'s current role emphasizes project-based learning in a small setting, often one-on-one with students, but she also highlighted other instances where she could apply PjBL in small, self-contained classrooms. None of the participants mentioned traditional quizzes and tests as assessment options.

S.F. and M.G. discussed their experiences collaborating with other professionals on PjBL. M.G. said she helped "the teachers modify and accommodate the lessons." S.F. discussed her experiences working with her classroom teams on behavioral challenges, which led them to use PjBL to boost student engagement and reduce non-preferred behaviors. These instructional features formed the foundation for the teachers' experiences, adaptations, and perceptions of impact.

Research Findings and Themes

Data from participant interviews and artifacts were analyzed multiple times through various coding transcripts and materials sessions to identify words and phrases that helped answer the research question aligned with the theoretical frameworks. Three themes emerged

from the data: supporting student learning, impact on students, and impact on teachers. Each theme includes sub-themes, as shown in Figure 2.

Figure 3

Themes

Theme	Subthemes
Supporting Student Learning	-Building Trusting Relationships -Adaptation Strategies
Impact on Students	-Academic and Cognitive Growth -Student Agency and Ownership -Behavioral, Social, and Emotional Development -Engagement and Motivation
Impact on Teachers	-Enjoyment and Flexible Teaching Experiences -Workload and Group Management Challenges -Professional Growth and Reflection

Supporting Student Learning

This theme describes how teachers establish foundational relationships to support students with EBD in PjBL. It involves establishing trust, understanding students' backgrounds, and adapting instruction. The theme of supporting student learning emerged repeatedly throughout all three participants' data. This theme has two subcategories: building trusting relationships and adaptation strategies.

Building Trusting Relationships. This sub-category emphasized building rapport and trust with students by showing empathy, understanding their backgrounds and needs, and being

flexible in addressing their challenges. The participants described building trusting relationships as rooted in understanding students' backgrounds and needs. B.Q. explained, "It's very rewarding ... so really getting that rapport with students and getting to know them and know how far you can push them, or how far you can try and get them to do the work."

All participants discussed their students' backgrounds, often mentioning their gaps in knowledge or skills related to academics, behaviors, and social skills. M.G. explained, "They have so many emotional challenges … They may have learned it but just don't remember because of trauma." This understanding influenced how the participants worked with their students. As B.Q. shared,

> There'll be days where you're like, okay, we're clearly not gonna do any work today. But let's just keep them stable for today versus, you know, other days where you can be like, alright, we're good. We can probably knock out like a day and a half or two days of work, because they're locked in.

All three participants emphasized the importance of what M.G. described as building "strong relationships with students." She said this was crucial for success: "It's more about the rapport with the students." B.Q. added,

> Once you built a rapport with them, and kind of getting to know them and getting them out of a funk, because a lot of times they'll come in and put their head down. I'll have a quick chat with them, and then they'll be fine within 5 or 10 minutes.

The words trust and rapport-building appeared multiple times in the data. S.F. reflected on the lasting value of building relationships when discussing comments from former students, saying, "We had a lot of kids that would call in, you know, and check in and 'Miss, you know, I really miss you guys.'" When asked about PjBL's effects on her students, M.G. perceived student

trust as important in getting the students to engage in PjBL. “They trust you more. ‘She doesn’t think that I’m not willing to do this. She understands that I’m having a little bit of issues, and I’ve never learned this before.’”

Participants' knowledge of their students' backgrounds led to a deeper awareness of what students needed to succeed. This insight was reflected throughout the data. Participants described students' academic challenges, including difficulties with writing, understanding traditional material, memory, and connecting with the “traditional” curriculum, which often felt irrelevant. M.G. noted, “Traditional lecture-based instruction often fails to meet their needs.” Social skills deficits and behavioral challenges were also common. M.G. shared, “Some of our kids even now, they’re in twelfth grade and they don’t know how to talk to each other.” B.Q. added, “A lot of the social stuff is really hard.” He further explained, “setting them up for success is super important.” M.G. added, “So we try our best, all of us, to accommodate whatever they need.” Building trust and understanding students’ backgrounds laid the foundation for the instructional adaptations discussed in the next section.

Adaptation Strategies. Participants discussed how they implemented adaptation strategies, including modifications to content, assessments, instructional methods, and grouping practices. Both S.F. and M.G. noted that adaptations are integrated into PjBL. S.F. explained, “I do think project-based learning kind of has its own adaptations built-in,” while M.G. described it as “actually scaffolding.” S.F. added,

> Well, with the integration of vocabulary and terminology, it allowed for them to actually gain the understanding of the actual physical manipulation tied in with the vocabulary, whereas when it was just vocabulary and just rote memorization, it didn’t allow for them to accommodate to those needs.

Beyond these statements, participants described adaptations such as making content more relevant, creating more hands-on experiences, and using grouping strategies. M.G. explained how she adapted assignments from the content teacher: “I scaled the complexity and structure to match their developmental level.” Similarly, B.Q. mentioned during planning that he would “start where they’re at.” S.F. discussed how her team used behavior data to adapt their instructional approaches and ultimately implement PjBL.

> So, with realizing with the FBA [functional behavior assessment] is that students were engaging in the behaviors of concern when they were doing less preferred activities or when they wanted to engage with their peers or adults. We kind of came to the idea that finding things that were more preferred, yet still achieved educational outcomes that we desired, kind of just benefited us both.

She also explained how they used learning style data to include more “experiential learning.” Additionally, the data from her lesson plan artifacts contained extensive differentiation strategies for struggling and advanced learners. In her lesson plan on angle exploration, students were asked to identify and classify different angles, measure, solve for unknown angles, and ultimately “design and execute precise movement sequences for the Sphero Indi robot by manipulating tile orientation” in a collaborative, problem-solving activity. S.F. provided options for struggling learners, including directions to “provide pre-drawn angle templates for students to place their tiles on for measurement practice,” and work “with smaller groups and provide more direct scaffolding and immediate feedback.” For more advanced learners, she included differentiation options such as allowing students to “research real-world robotics movements that rely on precise angles and try to replicate them.”

M.G. also discussed how she adapted on an individual level. She shared an example of a student who needed support with a literature assignment on Metamorphosis:

> She had to make it her own, so it was very hard for this cadet to understand how to do this, because she's never read anything like this… and I'm like, 'okay, listen. You know Coraline? Do you know this? And yeah, I know Coraline.' And then she made her own project, you know, with the modified instructions. She made her own projects, and they were awesome, like she made totally what the assignment wanted, but with project-based learning.

Both S.F. and B.Q. discussed adjusting group arrangements to support effective learning experiences. B.Q. noted,

> I did have to separate a few kids every once in a while … I still try to as much put them together, at least for a few seconds and be like, you're gonna have to learn how to work with people in the real world. So, I'm not gonna put you all on a two-week-long project, but if there's a half-day, half-class-period project, I might group you together.

He also described assigning group roles to leverage students' strengths. These examples illustrate how teachers adapted project-based learning to meet students' needs, preferences, and developmental levels, highlighting the importance of building trusting relationships to support these adaptations.

Impact on Students

The second theme that emerged from the data was its impact on students. The data reflected teachers' perceptions of the outcomes of PjBL for students with EBD, including academic, behavioral, social, emotional, and motivational effects. This theme includes four

subcategories: academic and cognitive growth; behavioral, social, and emotional development; engagement and motivation; and student agency and ownership.

Academic and Cognitive Growth. This subtheme included participants' perceptions on improvements in critical thinking, creativity, content knowledge, and retention from engaging in PjBL. Participants shared several common descriptions of how students grew academically and cognitively. B. Q. described,

> A lot of the students that I work with are going from a lower level, and I feel like they're increasing more as a percentage … so I feel like yeah, between the academics and the social emotional stuff, it's percentage-wise, and seeing them grow is a lot more rewarding.

M.G. perceived that PjBL "helps the kids relax and do good," and she believed that PjBL was easier for students than traditional lessons. "I just remember one kiddo with doing a project. It was just so much easier for him to do that instead of reading. You know what I mean, because it just set them off." She added that PjBL allows for more relevant learning that is easier for students to "get and grasp." She found that "project-based learning fosters deeper learning, students not only master content, but also develop essential skills like communication, critical thinking, and collaboration … allowing them to excel academically and personally."

S.F. explained it "encourages creativity and problem-solving within the material constraints." Retention of information was also observed. S.F. noted,

> Well, with the integration of the vocabulary and terminology it allowed for them to actually gain the understanding of the physical manipulation tied in with the vocabulary, whereas when it was just vocabulary and rote memorization, it didn't allow for them to accommodate to those needs … So, being able to hit in multiple different ways of

> learning through the same lesson, gave them more capability to remember the facts and actually synthesize the information rather than just memorizing for short term memory. So I think it was more conceptual, conceptualized for the long term.

Collectively, these perspectives illustrate that project-based learning was perceived as an effective approach for supporting students' academic development, conceptual understanding, and critical thinking skills.

Behavioral, Social, and Emotional Development. Data from this theme included decreased challenging behaviors, growth in social skills and emotional regulation, and the development of more adaptive behaviors. All three participants described experiences where PjBL positively impacted student behaviors, social skills, and emotional development. They noted that students gained valuable communication, critical thinking, and collaboration skills. B.Q. observed, "Learning to work with other people ... really showed how well they could do with teamwork." M.G. added that PjBL supported social and emotional growth, "The social aspect of it, it's huge … project-based learning has been a powerful instructional approach that fosters engagement, builds confidence, and supports social-emotional growth." M.G. also perceived that PjBL helped "with the trauma" students experienced, and positively impacted behaviors. "The behaviors aren't usually there [during project-based learning lessons]," she explained, "by offering choices within the project, this boosted their sense of control and reduced anxiety, conflicts, or off-task behaviors as opportunities to teach teamwork, communication, and resilience."

S.F. talked about engagement in the learning: "I mean, I have kids that never got up and engaged with anybody, and I did not have that entire week. I did not have a single kid that stayed in their seat during my lessons." She further described an example of behavior improvement:

> I had a zero percent head down rate during that week too, because we were tracking that data with it, because we had, I mean, we probably had about a third of the kids that would frequently put heads down, and that week we literally had nothing.

S.F. lesson artifacts guided student reflection. In her lesson on angles using Indi Robots, students needed to reflect on "a challenge they faced and how they overcame it." She described how students reflected more on their work.

> Most of it was rubric-based when we did project-based learning, and they would be more reflective as well on what their outcomes were like. They would come back and kind of evaluate how they could have done better … It's allowing them to actually recognize their own capabilities.

Additionally, S.F. stated that students learned more replacement behaviors during PjBL instruction.

> In the past, a lot of them we seem to have seen had been reinforced for acting out, or not engaging appropriately, so when they were engaging during that [PjBL] instruction, we were able to reinforce them for the appropriate engagement.

These findings illustrate that project-based learning was perceived as an effective approach for promoting positive behavioral changes, supporting social-emotional development, and fostering adaptive skills.

Engagement and Motivation. The theme of engagement and motivation included increased student interest, sustained attention, and enthusiasm for learning, all driven by relevance, flexibility, and creativity in project-based experiences. Most data from all sources centered on student engagement and motivation. Participants reflected on how the relevant and

flexible nature of PjBL encouraged greater engagement and creative expression. B.Q. noted that the flexible timing helped students with inconsistent engagement.

> If they're not feeling it, or have a rough day for one day, they're not for lack of a better word, screwed. They're not like, they can kind of catch back up if they lock in, or they're feeling much better on a different day. So, in that aspect, it really helps.

He also discussed flexibility in creative expression: "It also gives them a little leniency with their artistic or ability." Artifacts from M.G. showed how students could engage in creative expression through flexible assessments, including artistic expression and poetry.

M.G. stated, "I began using PjBL because I saw how much more engaged the students became when their learning felt relevant and hands-on … it was real learning." Relevancy and real-world application also appeared in the artifacts from all participants, demonstrating how authentic projects supported sustained engagement. M.G. shared several examples of student artwork created in response to English literature content.

> It's taking their culture, and different cultures from around the world, and figuring out, "okay, what's different? What's the same?" … This cadet could not think of anything around the world that she could, you know, she's from a rural area. So I said, "Let's make it where you're at. You're at [school name], and you have all these different types of people you're with in your platoon."

M.G. shared a picture of the final project, which showed images of the two students and highlighted what was the same and different about their experiences. This adaptation to the assignment allowed the student to be engaged by connecting it to relevant experiences.

B.Q. shared the final project, a model of the International Space Station that his students were responsible for creating.

> I had each group create one small piece of it, and they had to work together. And in the last few days, they had to work together to kind of piece them together, kind of just like how the International Space Station was created with different countries working together.

The lesson on angles that S.F. shared began by "asking students where angles are crucial in the real world. Guide them to examples like architecture, sports (e.g., billiards, basketball shots), navigation, robotics, or engineering." Her lesson on buoyancy, volume, and displacement also started with an engaging question: "Why do some things float and other's sink?" Students were then asked to build a device to "float and carry as much 'cargo' as possible without sinking." In the reflection part of the lesson, students discussed how large ships float. S.F. reflected that "The project-based learning was more incentivizing for them. I think they had more desire to learn…giving us the opportunity to give them the education that they should have been receiving."

Student Agency and Ownership. This theme focused on students' increased autonomy and sense of control during PjBL. Several artifacts shared by M.G. demonstrated opportunities for students to take control of their learning, including sample projects and lesson plans with various assessment options. Several assignments offered students different assessment choices. A literature assignment asked students to create a poster highlighting the key components of a story, and students were given three options for a "symbolic, modern reinterpretation" of the story. The student artifact accompanying the lesson plan displayed the choices students made while adding more of their personal interpretation of the literature. One assignment adapted by M.G. initially focused on Hamlet and asked students to create a social media post depicting what

it would look like; however, the student had never heard of Hamlet before and struggled with written expression. M.G. described her response to the student's lack of background knowledge:

> You know, with a cadet or a student, or any person who has an IEP with emotional disturbance … you don't want that person to get frustrated … so with *Hamlet*, we're like, "No, don't worry about *Hamlet*. What book do you really think you could change and tell people about?"

M.G. helped find a story the student knew well, enabling her to use her strengths for the final project. "She loves art. She drew this really crazy looking picture," M.G. said. "They have ownership of their work." S.F. also mentioned, "I think there was more ownership with their learning. I think that they had more desire to learn."

B.Q. described that PjBL gave "them kind of onus on their learning, because they were able to more or less steer their learning a little bit as to how and why they were learning, so it gave them a little more purpose." He elaborated further:

> It gives them more ownership of their learning, which helps a lot of the time when those students were told, 'You need to memorize this or you need to do that.' It just clashes, it kind of just stops them dead in their tracks. So even though they are being told you have to do this project, there is kind of leniency with how they go about it … what designs they want to do and stuff like that, so that out of everything, I feel like that having ownership probably helped the most.

M.G. noted, "Watching them take ownership of their learning, researching, collaborating, debating, and even creating visual media convinced me that project-based learning was more than just a fun alternative." Participants' experiences demonstrated that project-based

learning contributed to a range of positive student outcomes, including cognitive growth, increased motivation, improved social and emotional skills, and greater agency.

Impact on Teachers

The final theme identified from the data was the impact on teachers. This theme describes how implementing PjBL affected teachers' professional experiences, including positive aspects, challenges, and opportunities for personal growth. It includes three subcategories: flexible teaching practices, workload and group management challenges, professional development, and reflective practices.

Enjoyment and Flexible Teaching Experiences. This sub-theme encompassed the sense of satisfaction, flexibility, creativity, and professional fulfillment that teachers gained from using PjBL approaches. The participants described their teaching experiences as enjoyable. When asked what effects using PjBL had on M.G., she said, "It's fun." She also spoke positively about lessons: "I love this one. It was a metamorphosis lesson."

The participants highlighted that it was rewarding and offered more opportunities for creativity. B.Q. repeatedly mentioned that it was "a lot more rewarding" and had "more pros than cons." He explained, "even though it's arguably more work than being a math teacher, it's more rewarding and more fun." He valued the flexibility: "It gave me more leeway for projects where I can kind of choose how we're going to go about teaching this broad concept or project … you have more flexibility." Using PjBL also allowed him to incorporate other skills from his background: "It kind of brought a lot of my skill sets into one thing rather than it just focusing on my one math part … a lot more of my skill sets were used in this." S.F. noted that PjBL provided more opportunities to reinforce replacement behaviors. Replacement behaviors are "a positive alternative behavior that serves the same function" (Iris Center, 2025b, para. 1) as the behavior

teachers want students to reduce. “When they were engaging during that instruction, we were able to reinforce them for the appropriate engagement.” Participants described project-based learning as a rewarding and effective teaching method that increased their professional satisfaction, encouraged creativity, and allowed for greater flexibility in instruction.

Workload and Group Management. Participants shared difficulties related to managing group dynamics and increased preparation demands. While most of their experiences were positive, they also faced some challenges. B.Q. mentioned the difficulties of managing the group: “I did have to separate a few kids every once in a while.” He explained how uneven participation sometimes led to tension.

> It does affect the other students. And they’re like, “Well, I’m doing all the work for this day. Why are they not doing anything? Why, I better get more credit than them.” They’re very fixated on their grade and what that looks like, which, as in education, it’s hard to tell them, “Don’t worry about the grade.” Because it’s super important.

Considering the social challenges, he added, “It’s the learning that’s more important to them, and getting them to realize also that treating everyone the same isn’t the best option.

S.F. was the only participant who described truancy as a challenge: “I think that truancy is probably the biggest downfall, is making sure that they were able to catch up on whatever they were missing.” She elaborated, “We had times where students were sometimes hospitalized … sometimes it’s just for a day, but sometimes it’s for much longer, and you can’t just make up that kind of stuff really quickly.”

Two participants noted that planning was more work. S.F. explained, “There is definitely more preparation with it. So being able to gather all of the artifacts and materials needed prior to the lesson obviously is a huge component of it." B.Q. added, “I feel like the hardest part of my

job is coming up with the projects and the best ways to kind of convey that information and get them to experience that." Despite these challenges, both participants emphasized the benefits. As B.Q. summarized, “Even though it’s arguably more work than being a math teacher, it’s more rewarding and more fun.” S.F. added, “It’s more work to kind of prepare, but less during the instruction, because the kids drive so much more of their own learning.”

Professional Growth and Reflection. The final subtheme included teachers’ reflections on how PjBL expanded their instructional perspectives, shifted their mindsets about teaching and learning, and contributed to their ongoing professional development. All three participants reflected on their professional growth from implementing PjBL. B.Q. explained, “It definitely broadened my viewpoint on learning and education. It shifted my mindset a lot more from, you know, just getting them to memorize certain skills or things, to basically designing a whole experience for them.”

S.F. described how she would improve her PjBL experience: “As I’ve been learning more about introducing vocabulary and terminology in the actual experiential learning and I do feel that, that would actually provide a much more rounded education for the kids.” She also noted, “For the positive, I definitely feel like it’s taught me how to prepare better. It’s more work to kind of prepare, but less during the instruction because the kids seem to drive so much more of their own learning." She connected her PjBL experiences to broader shifts in educational standards, explaining, “I hope that everybody sees the positivity that can really come out of it.” Despite its workload and management challenges, participants’ reflections suggest that project-based learning was both professionally rewarding and transformative.

Summary

The themes identified through an iterative analysis described the experiences and perspectives of educators using PjBL with students with emotional and behavioral disabilities. Participants valued building relationships and understanding their students' needs, influencing their adaptation strategies. Notably, two participants mentioned that PjBL included adaptations within the learning experiences. Participants observed positive growth in their students' academic, cognitive, behavioral, social, and emotional skills. Their students were more engaged and motivated, with increased flexibility and ownership of their learning. The participants viewed their PjBL experiences positively, finding it rewarding, flexible, and worth the challenges of group work and workload. Finally, they reflected that their experiences broadened their understanding of learning, highlighted ways to improve, and inspired hope for other educators to recognize the value of PjBL.

Chapter V. Conclusions and Recommendations

This collective case study revealed teachers' experiences and perceptions using project-based learning (PjBL) with students with emotional and behavioral disorders (EBD). Students with EBD are at risk for many adverse outcomes (Chen et al., 2021; Didion et al., 2018; Kumm et al., 2021; Enis et al., 2018; Hutchins et al., 2020; Lipscomb et al., 2017; Mathews et al., 2021; National Center for Education Statistics, 2024; Offerman et al., 2022; Smith et al., 2018). Applying evidence-based practices with fidelity in this group has been difficult because of their complex needs (Criss et al., 2024; Scheibel, 2023; Sutherland et al., 2019). PjBL is an instructional approach that shares features with high-leverage practices (HLP) (Wertz & Mulcahy, 2024) and has seen success with students without disabilities (Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Gomez-Pablos et al., 2017; Revelle et al., 2020; Samsudin et al., 2020; Sormunen et al., 2019). The limited research on PjBL with students with EBD shows potential (Foster & Shriner, 2023; Taylor et al., 2024).

The study was focused on the following research question: What are the experiences and perspectives of teachers using PjBL with students with EBD? High leverage practices (HLPs) and the theoretical foundations of culturally responsive pedagogy and practices (CIPP) served as frameworks for this study. Three participants were interviewed, and artifacts related to their PjBL experiences were shared. Three themes emerged from the data: supporting student learning, impact on students, and impact on teachers. This chapter discusses findings, limitations, and implications for future research.

Discussion of the Findings

Participants included a special education teacher at a military academy, an eighth-grade STEM teacher from a central Pennsylvania school district, and a former special education

teacher from Pennsylvania teaching in self-contained settings. Data from participant interviews and artifacts were analyzed multiple times through various sessions of coding transcripts and materials to identify words and phrases that helped answer the research question in line with the theoretical frameworks. The findings revealed the participants' value in building relationships to understand how to meet their students' needs, their perceptions of the positive impact on the whole student, and the effect using PjBL had on them as educators.

Supporting Student Learning

The findings on supporting student learning described how teachers establish foundational relationships to support students with EBD in PjBL. This involved establishing trust, understanding their backgrounds, and adapting instruction. This theme aligned with several principles within CIPP and HLPs. Data analysis revealed that participants emphasized building rapport and trust with students by showing empathy, understanding their backgrounds and needs, and being flexible in addressing student challenges.

Participants emphasized the importance of building “strong relationships with students.” M.G. described this as essential for success: "It’s more about the rapport with the students.” S.F. reflected on the lasting value of building relationships when discussing comments from former students, saying, “We had a lot of kids that would call in, you know, and like check in and be like, Miss, you know, I really miss you guys.” These findings align with the CIPP principle of student-teacher relationships, emphasizing "caring, relationships, interaction, and classroom atmosphere" (Brown-Jeffy & Cooper, 2011, p.77). The findings also support the principle of identity and achievement within CIPP, which encourages awareness of students’ identities and perspectives, including honoring student assets (Brown-Jeffy & Cooper, 2011). The participants

described building trusting relationships as rooted in understanding students' backgrounds and needs, which led to a deeper understanding of what students required to succeed.

Participants discussed how they implemented adaptation strategies, including modifications to content, assessments, instructional methods, and flexible grouping arrangements. The CIPP principle of equity and excellence aligned with this subcategory. This principle encompasses the concepts of "dispositions, incorporation of multicultural curriculum content, equal access, and high expectations" (Brown-Jeffy & Cooper, 2011, p. 74). Adaptations included making content more relevant, creating more hands-on experiences, and using grouping strategies. M.G. described how she adapted assignments from the content teacher: "I scaled the complexity and structure to match their developmental level." Similarly, B.Q. mentioned during planning that he would "start where they're at." S.F. discussed how her team used behavioral data and learning style data to adapt instructional approaches and implement PjBL to meet the needs of students requiring more experiential learning. Additionally, the data from her lesson plan artifacts showed extensive differentiation strategies for struggling and advanced learners. These strategies aligned with the CIPP principle of developmental appropriateness, guiding teachers to tailor instruction based on learning styles and psychological needs (Brown-Jeffy & Cooper, 2011). They also align with HLP 13, 15, and 17, (see Appendix M) emphasizing the importance of adapting curriculum tasks and materials (HLP 13), using scaffolding to support learning (HLP 15), and employing flexible grouping to address diverse needs (HLP 17). (Lancaster et al., 2024).

Teaching the whole child is an additional CIPP principle aligned with the findings. This principle guides teachers to look beyond the surface-level needs of students and "remember the needs of the total child" (Brown-Jeffy & Cooper, 2011, p. 76). Teachers need to consider

students' cultural backgrounds, beliefs, and priorities of families and how these intersect to empower students to succeed (Brown-Jeffy & Cooper, 2011). Although family background was not part of the data, all participants mentioned aspects of meeting student needs beyond academics. These included addressing social needs by adjusting group roles and makeup and leveraging their strengths in decision-making. Social skills deficits and behavioral challenges were common. As B.Q. explained, “setting them up for success is super important.” M.G. added, “So we try our best, all of us, to accommodate whatever they need.” These examples demonstrate how teachers adapted project-based learning to meet students’ needs, preferences, and developmental levels.

Notably, two participants felt that adaptations were naturally part of the PjBL process. S.F. explained, “I do think project-based learning kind of has its own adaptations built-in,” while M.G. described it as “actually scaffolding.” S.F. added,

> Well, with the integration of vocabulary and terminology, it allowed for them to actually gain the understanding of the actual physical manipulation tied in with the vocabulary. In contrast, when it was just vocabulary and just rote memorization, it didn’t allow for them to accommodate to those needs.

The theme of supporting student learning emphasizes how participants combined relationship building with instructional practices to address the complex needs of students with EBD. Establishing trusting relationships provided the foundation for effective instructional strategies. The combination of relationship building and adaptations closely aligns with CIPP (Brown-Jeffy & Cooper, 2011). It is reflected in HLPs 13, 15, and 17 (see Appendix M), highlighting adapting materials, scaffolding, and flexible grouping (Lancaster et al., 2024). These

strategies fostered classroom environments where students could engage, participate, and grow academically and socially.

Impact on Students

The second theme that emerged from the data was its impact on students. The findings reflected teachers' perceptions of the outcomes of PjBL for students with EBD, including academic, behavioral, social, emotional, and motivational effects. This theme includes four subcategories: academic and cognitive growth; behavioral, social, and emotional development; engagement and motivation; and student agency and ownership. Data revealed that participants perceived growth and impact in all these areas. B.Q. described academic growth as "I feel like they're increasing more as a percentage," referring to students' academic success. Participants perceived a positive impact on student critical thinking skills, creativity, content knowledge, and retention, which were also positively affected by PjBL. As S.F. explained, "being able to hit in multiple different ways of learning through the same lesson gave them more capability to remember the facts and actually synthesize the information rather than just memorizing for the short-term memory." While there is no current research on academic growth with PjBL and students with EBD, these patterns reflect similar trends identified in studies with PjBL and students with learning disabilities (Lu et al., 2022) and students who were deaf, blind, or intellectually disabled (Eldiva & Azizah, 2019). Research has also shown positive academic growth for students without disabilities (Aydin & Mutlu, 2023; Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Gomez-Pablos et al., 2017; Krajcik et al., 2023; Revelle et al., 2020; Saavedra et al., 2021; Schneider et al., 2022; Sormunen et al., 2019; Suryanti et al., 2024; Wang et al., 2022; Zhang et al., 2023).

Participants also perceived a positive impact on student behavior, social skills, and emotional development. They emphasized that students developed important communication, critical thinking, and collaboration skills. B.Q. said, “Learning to work with other people ... really showed how well they could do with teamwork.” M.G. noted that PjBL supported social and emotional growth and helped students process trauma. “The behaviors [aggression, shutting down, task refusal] aren’t usually there," she explained, “by offering choices within the project, this boosted their sense of control and reduced anxiety, conflicts, or off-task behaviors as opportunities to teach teamwork, communication, and resilience.” S.F. shared how one PjBL unit positively influenced students who were usually more avoidant of work and helped them improve their replacement behaviors and “actually recognize their own capabilities.”

The findings demonstrate that PjBL can align with HLP 9, (see Appendix M) enabling teachers to teach social behaviors (Lancaster et al., 2024) and support the CIPP practice of educating the whole child, emphasizing the importance of addressing students' overall needs beyond academics (Brown-Jeffy & Cooper, 2011). These findings are consistent with studies indicating that teachers observed improved social skills in students with disabilities when PjBL was implemented (Purnomo et al., 2022). Additionally, the results in this theme align with previous research demonstrating enhanced student behavior in students with EBD who were instructed using PjBL (Foster & Shriner, 2023; Taylor et al., 2024).

Data also showed that participants believed their students were more engaged and motivated to learn, influenced by relevant content, flexibility, and opportunities for creative expression. Participants explained how the adaptable nature of PjBL supported higher engagement levels, especially for students who struggled to stay focused during traditional lessons. Artifacts shared by M.G. demonstrated how students could express themselves

creatively through flexible assessments, including artistic work and poetry. She said, “I began using PjBL because I saw how much more engaged the students became when their learning felt relevant and hands-on … it was real learning.”

Relevancy and real-world application emerged as factors contributing to engagement across all data sources. S.F. reflected that “Project-based learning was more incentivizing for them. I think they had more desire to learn … giving us the opportunity to give them the education that they should have been receiving.”

These findings align with previous research on PjBL with students with EBD. Foster and Shriner (2023) found that teachers in a residential school also perceived increased engagement when using PjBL with students with EBD. When teachers intentionally consider students’ needs and learning styles and adjust their instructional approaches, as described in the principle of developmental appropriateness (Brown-Jeffy & Cooper, 2011), students are more likely to be engaged and motivated to learn. PjBL is one method to foster this engagement, as the findings showed that providing relevant, hands-on experiences and flexible structures helped students participate more actively and consistently, aligning with HLP 18 (see Appendix M), which highlights strategies that promote sustained involvement in learning (Lancaster et al., 2024). These experiences support research emphasizing the role of PjBL in fostering engagement and reducing behaviors among diverse learners.

The final perceived impact on students was their increased autonomy and sense of control. Findings included statements and artifacts showing a variety of student choices for instruction and assessment. Students communicated their learning through group work, art, poetry, and problem-solving presentations. Throughout the data, student ownership was consistently described as one of the most beneficial aspects of PjBL for students with EBD. As

M.G. stated, “Watching them take ownership of their learning, researching, collaborating, debating, and even creating visual media convinced me that project-based learning was more than just a fun alternative.”

Like the engagement and motivation subtheme, fostering student ownership aligns with HLP 18 (see Appendix M), which emphasizes using strategies to engage students in active participation (Lancaster et al., 2024). Providing opportunities for choice and autonomy also supports high expectations, as teachers encourage students to make decisions about their work instead of relying on predetermined instructions. This approach aligns with the CIPP practice of equity and excellence, emphasizing high expectations for students and empowering learners to succeed (Brown-Jeffy & Cooper, 2011). Prior research also found that teachers perceived allowing students to develop ideas and make choices as increasing engagement (Foster & Shriner, 2023). These findings regarding the impact on students show how PjBL can be an empowering experience for students with EBD, leading to positive outcomes across both academic and non-academic areas.

Impact on Teachers

The final theme identified from the data was the impact on teachers. The findings revealed teachers’ perceptions of how implementing project-based learning affected their professional experiences, including positive aspects, challenges, and opportunities for personal growth. Participants described project-based learning as a rewarding and effective instructional approach that increased professional satisfaction, encouraged creativity, and offered more instructional flexibility. B.Q. explained that despite the increased workload, “it’s more rewarding and more fun.” The participants appreciated the flexible nature of PjBL for teachers, whether that

involved the instructional choices they could make or the addition of other teaching skills. S.F. added that it allowed her to include more opportunities to teach students replacement behaviors.

These findings are similar to other studies in which teachers reported positive perceptions of using PjBL with students with EBD (Taylor et al., 2024). Participant reports of flexibility in their instruction align with the principles of CIPP regarding the variation of needs and teaching styles based on developmental appropriateness (Brown-Jeffy & Cooper, 2011).

While most data indicated positive perceptions and experiences, participants shared difficulties managing group dynamics and increased preparation demands. The social challenges emphasized the need for intentional groupings. Participants reflected on these challenges, and their experiences highlighted the importance of building relationships and classroom communities, as seen in the CIPPs of equity and excellence and student-teacher relationships. B.Q. mentioned the difficulties of managing the group, but added that it required him to deliberately set up successful groups or support those who needed more help. He also described fostering empathy and relationships within the classroom community when other students expressed frustration over classmates who were not participating. Considering these social challenges, he added, “It’s the learning that’s more important to them, and getting them to realize also that treating everyone the same isn’t the best option.”

The findings showed that participants believed PjBL required more planning and preparation. Despite this, they continued to feel that it was worth the effort. Their enjoyment of the work aligns with prior research, where teachers reported increased job satisfaction using PjBL with students with EBD (Taylor et al., 2024).

The final impact revealed by the findings was the professional growth of the participants. Data showed how PjBL expanded their instructional perspectives, changed their mindsets about

teaching and learning, and supported their ongoing professional development. Their reflections and growth align with part of HLP 6 (see Appendix M), where teachers are asked to analyze their instructional practices and adjust as needed (Lancaster et al., 2024). The findings indicated professional growth in understanding learning, a desire to continue learning more, and ideas for improving instructional practices. S.F. noted that she would change how she introduces vocabulary in the future and felt that using PjBL helped her improve her lesson planning. Despite the workload and management challenges, participants' reflections suggest that project-based learning was professionally rewarding and transformative. "I hope that everybody sees the positivity that can really come out of it."

Limitations

Several limitations should be considered when interpreting the findings of this study. The participants implemented PjBL differently, reflecting the differences in their instructional contexts, student populations, and professional roles. This variability limits the consistency of comparisons across cases. While M.G. had experience planning full PjBL experiences in small, self-contained classes, her experiences at the time of the study were more aligned with adapting content and assessments from the content teacher at the military academy. B.Q. implemented PjBL in an inclusive setting, and S.F. in a self-contained emotional support setting. Additionally, when considering the elements of high-quality project-based learning, the data did not clearly represent all elements (The Buck Institute for Education, 2018). For example, the element of a public product appeared limited to peers in their classrooms or other teachers, except for B.Q., who noted that students collaborated across classes during one unit. It was also unclear how much emphasis was placed on student reflection outside of S.F.'s statements.

The participants held different professional roles. During this study, M.G. was a special educator supporting content teachers at a military academy, B.Q. was an eighth-grade STEM teacher in an inclusive classroom, and S.F. was a special educator working with students in a self-contained classroom. Although their diverse experiences provided valuable and diverse perspectives, the small sample size means the findings may not be transferable to other types of instructional settings.

My professional connection with two of the participants was a limitation. Although I had never met or interacted with M.G. before this study, she worked for the organization I was employed by at that time. We did not work in the same school or department within the organization, but this should be noted as a potential limitation. Additionally, I briefly interacted with S.F. during a statewide networking session before this study. These prior relationships could have influenced the study's results. There may have been potential bias in data collection, whether participants may have responded more positively or found it more difficult to feel anonymous or completely honest. It could also have influenced how I interpreted their experiences.

The classrooms described by the participants included students with various disabilities. All three participants mentioned they taught students with a range of different disabilities. Although the informed consent process and interviews emphasized the need to focus specifically on students with emotional disturbance, their recollections and examples may have included experiences with students with other disabilities.

The study's data were limited to interviews and artifacts. Artifacts included student work from two participants and lesson plans from two participants. No artifacts related to participant reflection or lesson materials were included. This data might have been influenced by recall bias

or participants' desire to present their experiences positively. A lack of observations may have been a limitation. Observations would have provided an additional opportunity to triangulate the data from interviews and artifacts by offering a different perspective on implementing PjBL with students with EBD.

This study focused on teacher experiences and perspectives but lacked the perspectives of students, families, and administrators. Including these perspectives would have provided a more complete understanding of how PjBL affected students with EBD. It could also have revealed additional insights about implementation challenges, student experiences, and systematic factors that influenced the use of PjBL with students with EBD.

Finally, my interpretations of the data may have been shaped by my professional experiences and beliefs about PjBL. I have professional experience using PjBL with students with EBD in self-contained settings. My recollection of these experiences is that PjBL is an effective way to teach students with EBD. Although I reflected on my biases and experiences before interviews, data analysis, and writing, they may still have influenced my report.

Implications

As mentioned in Chapter I, research on using PjBL with students with EBD is limited. This study's results add to the existing research in this area. Teachers' perceptions and experiences with PjBL were based on their understanding and student-teacher relationships. They perceived a positive impact on students' academic and non-academic areas. Although challenges related to group dynamics and planning were present, the findings indicated that participants believed PjBL was an effective strategy for students with EBD.

Participants perceived a positive impact on students. The findings showed that the teachers in this study believed PjBL positively affected their students. Educators and school

leaders can use these findings to advocate for more PjBL with students with EBD in inclusive and self-contained settings. PjBL could benefit other students beyond this study, as well as teachers. Educational leaders and administrators should provide additional professional development on PjBL, identify resources to help teachers reduce workload, and allocate more planning time for educators using PjBL to help manage workload demands. Additionally, professional development is recommended to include specific training on integrating social skills instruction and flexible grouping strategies to help teachers address common challenges when implementing PjBL with students with EBD. Regional organizations, such as intermediate units, should also consider coordinating a professional learning community for educators across their region to coordinate, collaborate, and learn from each other.

Recommendations for Future Research

As noted in previous chapters, research on PjBL with students with EBD is limited. Future studies should explore the impact and application of PjBL with students with EBD. The participants perceived a positive effect in this study, and S.F. cited behavior data for one unit; however, this can be expanded into a quantitative study that directly measures PjBL's perceived impact on student academic achievement, social skills, and behaviors. A quantitative study is recommended to measure student achievement through a curriculum-based assessment. Additionally, conducting a quantitative study to examine pre- and post-intervention behavior data for students participating in PjBL would provide valuable insights into its impact on behavioral outcomes. This approach could also be applied using a social skills assessment to examine the impact of PjBL on students' social development.

As mentioned in the limitations, this study did not include classroom observations of teachers. A recommendation is to conduct a case study that involves observing classes where

PjBL is used with students with EBD. In such research, it would be beneficial to have participants engage in high-quality PjBL experiences, using a rubric to ensure consistency if conducting a collective case study. A qualitative case study would build on the research presented here. Additionally, qualitative research could explore students' perceptions of PjBL to better understand its impact.

Finally, the long-term effects of PjBL on these students should also be explored. A mixed-methods longitudinal study tracking students over several years could determine if improvements in academic achievement and social skills last beyond the PjBL experience. Tracking students with EBD over multiple years could also show whether PjBL reduces adverse outcomes, including dropout rates (National Center for Education Statistics, 2024) and isolation from typical, same-aged peers (Office of Special Education Programs, 2020).

Conclusion

Students with emotional and behavioral disorders face various negative outcomes, including lower academic achievement and behavioral challenges (Chen et al., 2021; Smith et al., 2018), higher dropout rates (National Center for Education Statistics, 2024), and social isolation from their typical, same-aged peers (Office of Special Education Programs, 2020). While evidence-based practices are available to reduce these risks, consistent implementation remains difficult (Farley et al., 2023; Smith et al., 2018; Sutherland et al., 2019). Project-based learning (PjBL) is a promising instructional approach because it aligns with many high-leverage practices that support students with disabilities (Wertz & Mulcahy, 2024). Research with students without disabilities has shown positive results in academics (Aydin & Mutlu, 2023; Culclasure et al., 2019; Duke et al., 2021; Eldiva & Azizah, 2019; Febriana, 2017; Gomez-Pablos et al., 2017; Krajcik et al., 2023; Revelle et al., 2020; Saavedra et al., 2021; Schneider et

al., 2022; Sormunen et al., 2019; Suryanti et al., 2024; Wang et al., 2022; Zhang et al., 2023), as well as improvements in behavior and social-emotional learning (Culclasure et al., 2019; Febriana, 2017; Krajcik et al., 2023; Juniar et al., 2023; Lee et al., 2015; Samsudin et al., 2020; Wang et al., 2022). Studies involving students with other disabilities, such as speech, reading, and writing disabilities (Berkeley et al., 2019), learning disabilities (Lu et al., 2022; Sormunen et al., 2019), those who are blind, deaf, or intellectually disabled (Eldiva & Azizah, 2019), as well as students in inclusive settings with disabilities, including learning disabilities, giftedness (Hovey & Ferguson, 2014), autism, speech impairments, visual impairments, intellectual disabilities, physical disabilities, deafness, ADHD, and learners with "social and behavioral barriers" (Purnomo et al., 2022, p. 294), have reported similarly positive outcomes. Although specific research on PjBL with students with EBD is limited, existing studies suggest that this approach shows promise as an effective instructional method (Foster & Shriner, 2023; Taylor et al., 2024).

This collective case study revealed teachers' experiences and perceptions of using project-based learning (PjBL) with students with emotional and behavioral disorders (EBD). Criterion sampling was employed to select three participants who had used PjBL with students with EBD during the 2024-2025 school year. Data analysis, guided by the theoretical frameworks of high-leverage practices and culturally inclusive pedagogy and practices, revealed themes of supporting student learning, impact on students, and impact on teachers.

The findings of this study aligned with high-leverage practices (Lancaster et al., 2024) and culturally inclusive pedagogy and practices (Brown-Jeffy & Cooper, 2011), as well as with prior research on PjBL and students with EBD (Foster & Shriner, 2023; Taylor et al., 2024). Data analysis revealed that participants emphasized building rapport and trust with students by

demonstrating empathy, understanding their backgrounds and needs, and maintaining flexibility in addressing challenges. Their perceptions of student impact illustrated how PjBL can serve as an empowering experience for students with EBD, contributing to positive outcomes across both academic and non-academic domains. Teachers also reported that implementing PjBL enhanced their professional experiences and supported their growth. Despite the challenges of workload demands and group dynamics, PjBL was worthwhile for the participants and their students.

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Appendix A

Email to STEM Points of Contact

Dear (INSERT NAME HERE),

Hello! I hope you are having a good school year. I am conducting research for my dissertation on the experiences of teachers using project-based learning with students with emotional and behavioral disorders. If you are willing, please pass along this brief email if you know anyone who may teach project-based learning with this population? Thank you so much for your support! If you have any questions, please let me know! Thank you!

Sincerely,

Jen Sciacca

jennifer.sciacca@wilkes.edu

570-856-6035

Attached email:

Dear (INSERT NAME HERE),

Hello! I hope you are having a good school year. My name is Jennifer Sciacca, and I am a doctoral student at Wilkes University finishing my Ed.D. in Educational Research. I am conducting research for my dissertation on the experiences of teachers using project-based learning with students with emotional and behavioral disorders. I was given your contact information by a colleague who recommended that your experience and knowledge in this area would be valuable to include in my research. I value your perspectives and would like to know if you are interested in participating!

If you are interested, please fill out this brief Google form. https://forms.gle/h78xSQ6LJvaHm48G9 I will give you more details and information if you choose to participate. If you have any questions, please let me know! Thank you!

Sincerely,

Jen Sciacca

jennifer.sciacca@wilkes.edu

570-856-6035

Appendix B

Email to Potential Participants

Dear (INSERT NAME HERE),

Hello! I hope you are having a good school year. My name is Jennifer Sciacca, and I am a doctoral student at Wilkes University finishing my Ed.D. in Educational Research. I am conducting research for my dissertation on the experiences of teachers using project-based learning with students with emotional and behavioral disorders. I was given your contact information by a colleague who recommended that your experience and knowledge in this area would be valuable to include in my research. I value your perspectives and would like to know if you are interested in participating!

If you are interested, please fill out this brief Google form. https://forms.gle/h78xSQ6LJvaHm48G9 I will give you more details and information if you choose to participate. If you have any questions, please let me know! Thank you!

Sincerely,

Jen Sciacca

jennifer.sciacca@wilkes.edu

570-856-6035

Appendix C

Participant Survey

Survey to be completed on Google Forms. https://forms.gle/h78xSQ6LJvaHm48G9 See questions below.

Hello! Thank you for your interest in participating in research around teacher experiences using project-based learning with students with emotional and behavioral disabilities. Participants may include special education teachers in self-contained settings, general educators in inclusive classrooms, or special education teachers co-teaching in general education settings.

For this study, participants will be asked to participate in a one-on-one interview with the researcher and provide copies of project-based learning documents (such as lesson plans and materials).

Please complete this brief survey, and I will contact you to discuss the details!

1. Name
2. Contact email
3. Best phone number (optional)
4. Job Title
5. How many years of experience do you have teaching classes that include students with emotional or behavioral disorders?
6. Have you taught these students using project-based learning?
7. Have you taught a project-based learning unit in the last year with students with emotional and behavioral disorders?
8. Any other comments or questions?

Appendix D

Social Media Recruitment Post

Calling all teachers working with students with emotional or behavioral disorders (EBD)!

Are you using project-based learning (PjBL) in your classroom? I want to hear from you!

I'm conducting a study for my dissertation to understand teachers' experiences with using PjBL with students with EBD. Your participation could help shape future educational practices!

Eligibility:

Currently teaching classes that include students with EBD

Using project-based learning methods

Interested? Have questions? Fill out this brief Google Form, and I will contact you with more information!

Appendix E

Email for Informed Consent Meeting

Dear (INSERT NAME HERE),

Thank you again for your interest in participating in my research on teachers using project-based learning with students with emotional and behavioral disabilities.

I am emailing to schedule a meeting with you to review the informed consent letter attached to this email. I will also discuss the study in more detail and answer any questions. Let me know what day and time works best for you in the next two weeks to meet via Zoom or if you prefer another method of meeting. Thank you!

Sincerely,

Jen Sciacca

jennifer.sciacca@wilkes.edu

570-856-6035

Appendix F

Letter of Informed Consent

Wilkes University

School of Education

84 W. South Street Wilkes-Barre, PA 18766

Letter of Informed Consent

Title of Study: Teacher Experiences with PjBL and Students with EBD
Principal Investigator: Jennifer Sciacca
Phone: 570-856-6035
Email: jennifer.sciacca@wilkes.edu

Dear (NAME)

You are invited to participate in a research study conducted by Jennifer Sciacca as a graduate student to earn the Doctor of Education degree in Educational Leadership from Wilkes University. You should read the information below and ask questions about anything you do not understand before deciding to participate. If you agree to participate, you will be asked to sign this form, and you will be given a copy of the form.

Background and Purpose of the Study: Students with emotional and behavioral disorders (EBD) often face challenges in traditional educational environments. Research shows that project-based learning (PjBL) can lead to positive outcomes with other student populations; however, there is limited research on its application for students with EBD. This study aims to understand teachers' experiences using PjBL in classrooms with students who have EBD.

Study Procedures and Time Involvement: If you agree to participate, you will join Jennifer Sciacca for a recorded interview about your experiences using project-based learning when teaching students with emotional and behavioral disabilities. The interview is expected to last 45-60 minutes. Interviews will take place via Zoom and will be recorded for research purposes. You will also agree to provide documents, such as lesson plans and lesson materials (i.e., slideshows, worksheets, etc) related to a project-based learning unit you have taught.

Benefits and Risks: There are no direct benefits for participants in this study. The results of this study may inform further research and practice. The risks of participating in this study are no greater than those encountered in normal daily life.

Confidentiality: Once this consent form is signed and returned, it will be stored in a password-protected folder on a secure cloud drive that will be physically locked from public access. Once

an interview is scheduled, each participant will be asked to choose a pseudonym during the recorded interview.

Participant's Rights: Your participation is voluntary, and you have the right to withdraw your consent or discontinue participation at any time without penalty or loss of benefits to which you are entitled. You can choose not to participate. There is no future compensation or payment to participants as a result of this study.

Contact Information: If you have any questions or concerns about this research study, contact the principal investigator, Jennifer Sciacca (570-856-6035), jennifer.sciacca@wilkes.edu or Dr. Paul Reinert (570-417-6447) paul.reinert@wilkes.edu, who is the faculty member supervising this research.

If you have questions, concerns, or feel your rights have been violated as a research participant, you may contact the chair of the Wilkes University Institutional Review Board (IRB) at irb@wilkes.edu.

Statement of Consent: I have read the above information and agree to participate.

Signature _____ **Date:** _____

Name (Print) _____

☐ I agree to be audio-recorded/video-recorded/photographed.

☐ I do not agree to be audio-recorded/video-recorded/photographed.

Signature _____ **Date:** _____

Name (Print) _____

Appendix G

Pre-Interview Email

Hello!

I'm looking forward to our meeting on (DATE AND TIME). Below are some of the questions you will be asked so that you may look at them ahead of time. You do not need to prepare your responses.

1. Describe your background and professional experience in education.
2. Describe how you came to use project-based learning in your teaching.
3. Describe your experiences teaching students with emotional and behavioral disabilities (EBD) using project-based learning (PjBL).

This email also includes a link to the Zoom meeting. Please let me know if you have any questions or concerns before (DATE OF INTERVIEW)! Thank you again for your participation.

(INSERT ZOOM LINK HERE)

Sincerely,

Jen Sciacca

jennifer.sciacca@wilkes.edu

570-856-6035

Appendix H

Interview Protocol

Thank you for taking the time today to discuss your experiences using project-based learning in classes with students with emotional and behavioral disabilities. This interview will be recorded to assist in data analysis, and your information will be kept confidential. The research question that is guiding this pilot study is

> What are the experiences of teachers who have implemented project-based learning with students with emotional and behavioral disorders?

I prepared some questions that I previously sent to you ahead of time. As we go through the interview, there may be some follow-up questions relating to your experiences. I also may write down notes from time to time to assist in the process. If at any time you do not feel comfortable answering a question or wish to end the interview, please let me know. Do you have any questions before we start? (Begin recording after the interviewee responds with no more questions.)

Interview Questions

- Describe your background and professional experience in education.
    - Prompt
        - Describe your experiences teaching students with EBD.
        - Describe your current role or role when using PjBL with students with EBD.
- Describe how you came to use PjBL in your teaching.
    - Prompt:
        - How would you describe PjBL?

- Describe your experiences teaching students with emotional and behavioral disabilities using project-based learning.
    - Prompts:
        - Describe an example of a project-based learning unit you have taught with students with EBD.
        - What effects do you feel PjBL had on you as a teacher?
        - What effects do you feel PjBL had on your students?
        - Describe the benefits you and your students experienced during a PjBL unit.
        - Describe the challenges you and your students experienced during a PjBL unit.
        - Describe, if applicable, how PjBL met the needs of your students with EBD.
        - Describe, if applicable, how you needed to adapt PjBL to meet the needs of your students with EBD.
        - Is there anything else you want to add before we end the interview?

(STOP RECORDING)

Thank you again for contributing your knowledge to this pilot study. I will contact you throughout my data analysis if I have any follow-up questions or need clarification. Do you have any questions?

Appendix I

Citi Certification

CITI PROGRAM

Completion Date 01-Jun-2024
Expiration Date 01-Jun-2027
Record ID 43771468

This is to certify that:

Jennifer Sciacca

Not valid for renewal of certification through CME.

Has completed the following CITI Program course:

Social & Behavioral Research - Basic/Refresher
(Curriculum Group)
Social & Behavioral Research
(Course Learner Group)
1 - Basic Course
(Stage)

Under requirements set by:

Wilkes University

CITI
Collaborative Institutional Training Initiative
101 NE 3rd Avenue, Suite 320
Fort Lauderdale, FL 33301 US
www.citiprogram.org

Generated on 01-Jun-2024. Verify at www.citiprogram.org/verify/?w80b09eb5-db0a-4a9f-9382-385f769698a5-43771468

Appendix J

Email to Participants to Review Findings

Dear (INSERT NAME HERE),

Thank you again for participating in the interview for research, looking at your experiences in using project-based learning with students with emotional and behavioral disabilities.

After completing data collection, I was able to analyze the information and break it down into specific themes. I want to ensure that the data I have reported and analyzed accurately reflects your experiences. If you can, I would greatly appreciate you reviewing the attached draft of the findings and providing me with any feedback, additional information, or changes you feel are necessary. If you feel it is an accurate representation of your experiences, please feel free to let me know that as well!

Please let me know if you can complete the feedback process by (INSERT DATE TWO WEEKS FROM DATE OF EMAIL). Thank you again for your participation, and I look forward to hearing from you!

Sincerely,

Jen Sciacca

jennifer.sciacca@wilkes.edu

570-856-6035

Appendix K

Email to Colleague for Constructive Feedback

Dear (INSERT NAME),

Hello! I have recently concluded research looking at the experiences of teachers using project-based learning (PjBL) with students with emotional and behavioral disorders (EBD). These students often face challenges in traditional educational environments. Research shows that project-based learning can lead to positive outcomes with other student populations; however, there is limited research on its application for students with EBD. This pilot study aimed to understand teachers' experiences using PjBL in classrooms with students who have EBD.

I have attached a draft of my initial findings and the raw data. I would be grateful if you could review this and provide me with constructive feedback regarding whether the findings are plausible before I submit it for final review. If possible, I would need it by (INSERT DATE). Let me know if you have any questions. Thank you for your help!

Sincerely,

Jen Sciacca

jennifer.sciacca@wilkes.edu

570-856-6035

Appendix L

Email to Site Supervisor Requesting Access

Dear (INSERT NAME HERE),

Hello! I hope you are having a good school year. In addition to my work at the CAIU, I am a doctoral student at Wilkes University finishing my Ed.D. in Educational Research. I am conducting research for my dissertation on the experiences of teachers using project-based learning with students with emotional and behavioral disorders. I am contacting you to request permission to contact teachers in (SCHOOL NAME) who would qualify for this research. With your approval, I will send each qualifying teacher an initial email briefly explaining the research and what would be asked of them should they choose to participate. If you are willing to assist, I respectfully request that you provide the names and contact information of the teachers who you know use project-based learning with students with emotional and behavioral disorders. If you have any questions, please let me know! Thank you!

Respectfully,

-Jen Sciacca

Doctoral Student

Wilkes University

Appendix M

High-Leverage Practices Connected to Findings

Data Driven Planning	**Instruction in Behavior and Academics**
• HLP 6: Use student assessment data, analyze instructional practices, and make necessary adjustments that improve student outcomes (Iris Center, 2025a, para. 2).	• HLP 9: Teach social behaviors. • HLP 13: Adapt curriculum tasks and materials for specific learning goals. • HLP 15: Provide scaffolded supports. • HLP 17: Use flexible grouping. • HLP 18: Use strategies to promote active student engagement (Iris Center, 2025a, para 2).

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